

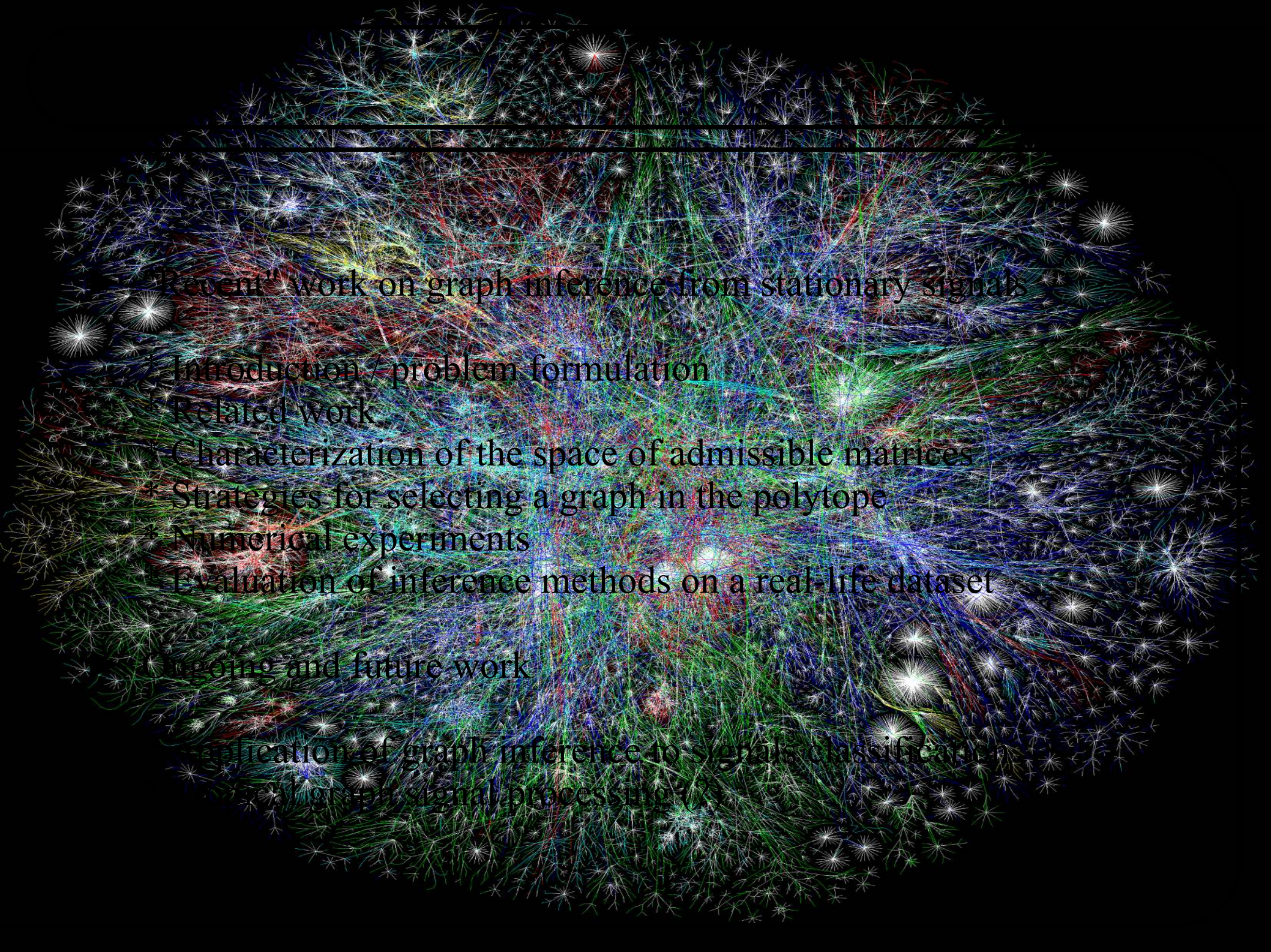
Characterization and Detection of Chaotic Processes from Observations of Stationary

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McGill



Recent work on graph inference from stationary signals

- 1. Introduction / problem formulation

- 2. Related work

- 3. Characterization of the space of admissible matrices

- * Strategies for selecting a graph in the polytope

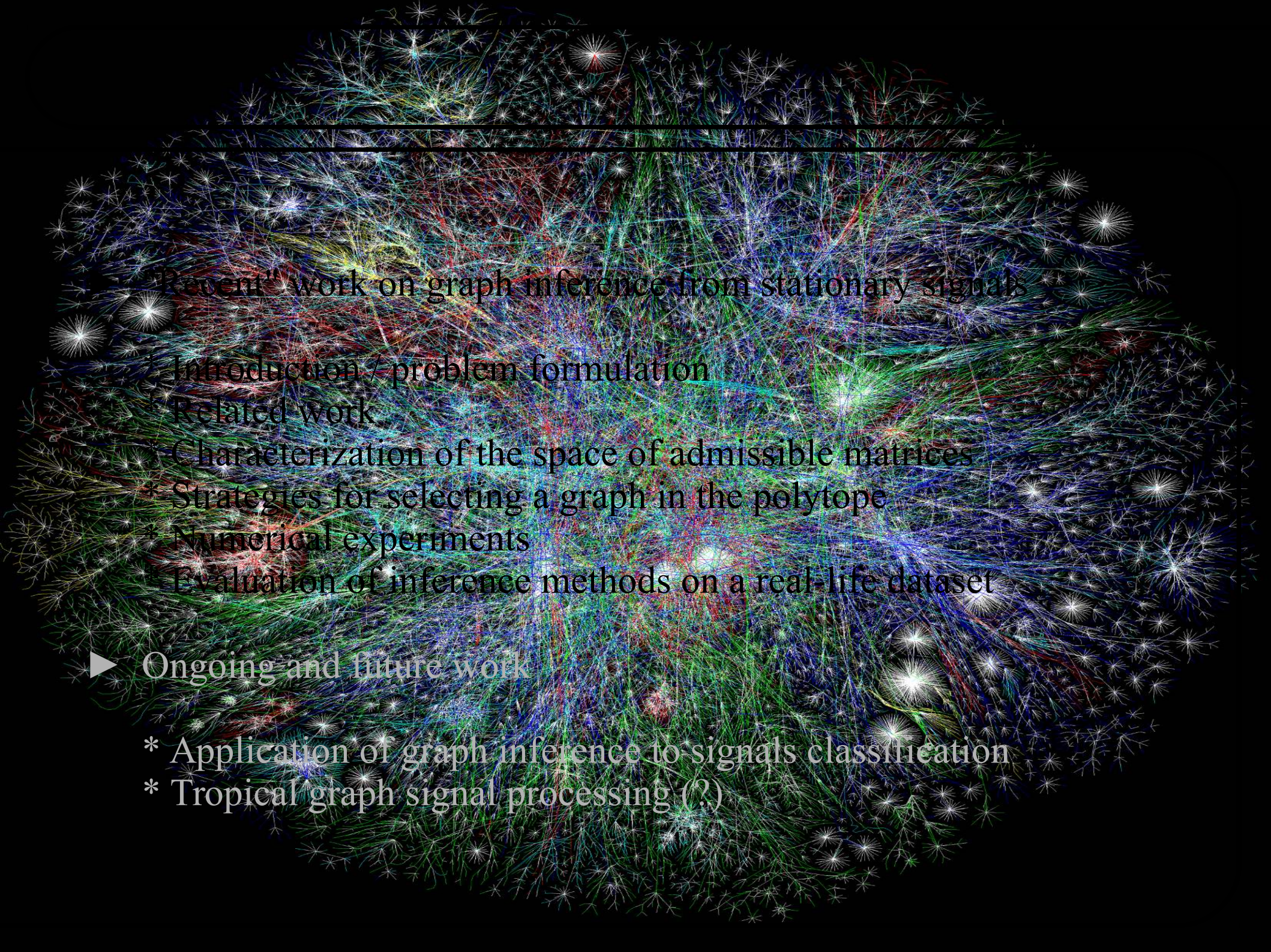
- * Numerical experiments

- * Evaluation of inference methods on a real-life dataset

- 4. Ongoing and future work

- 5. Application of graph inference to signals classification

- 6. Local graph signal processing (2)



Recent work on graph inference from stationary signals

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► Ongoing and future work

- * Application of graph inference to signals classification

- * Tropical graph signal processing (?)

► "Recent" work on graph inference from stationary signals

1 Introduction / problem formulation

* Related work

* Characterization of the space of admissible matrices

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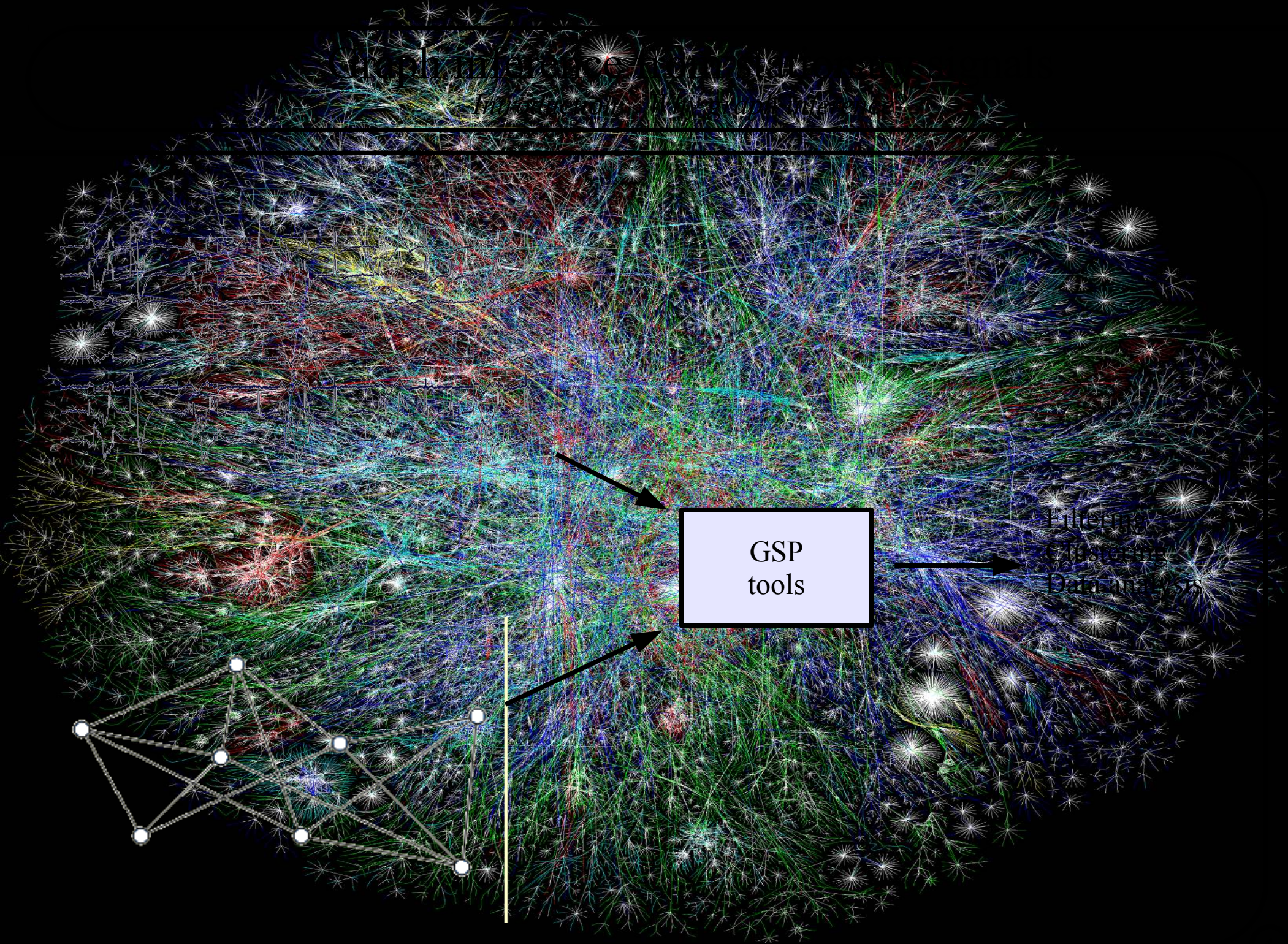
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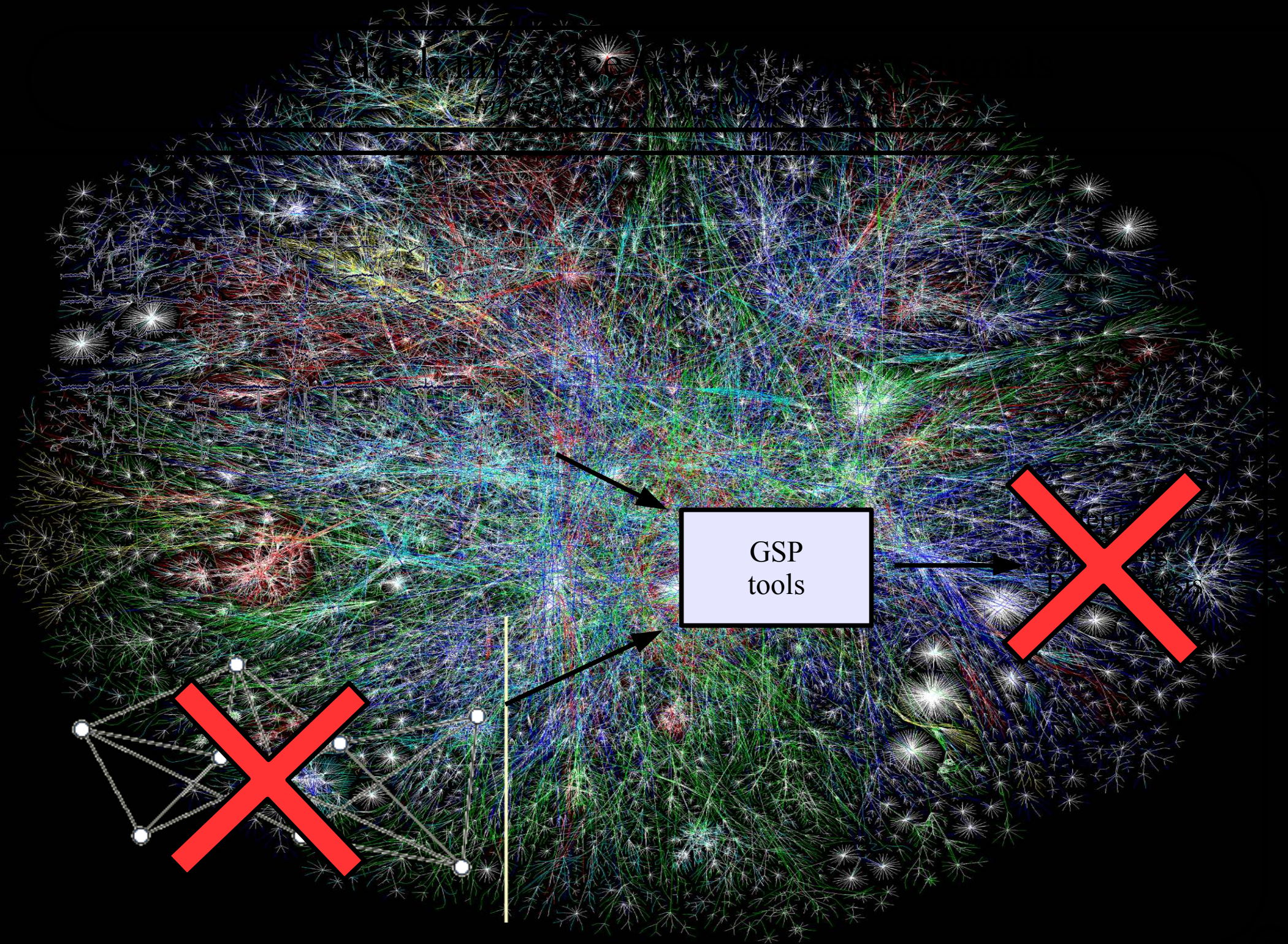
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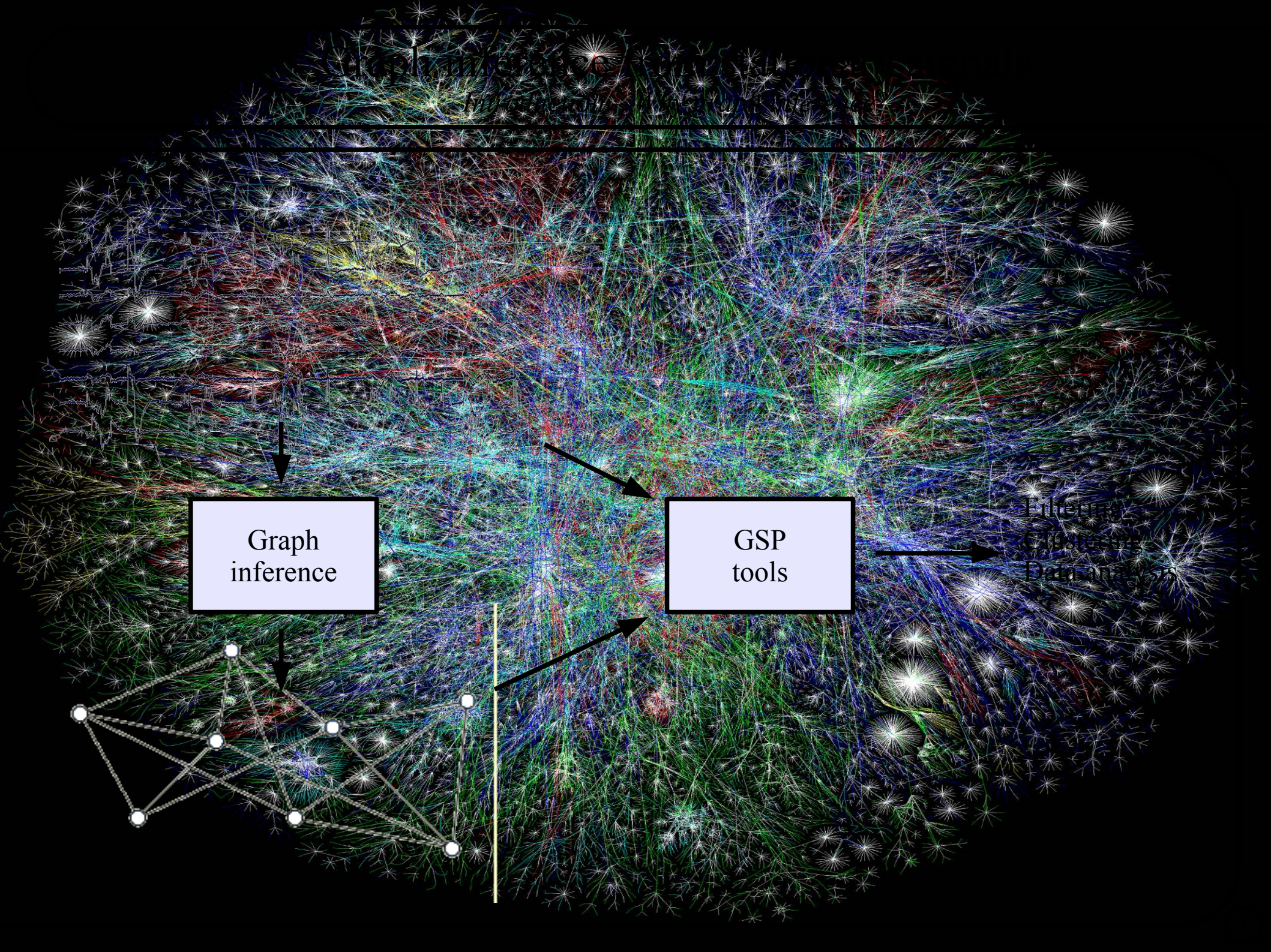
► Ongoing and future work

* Application of graph inference to signals classification

* Tropical graph signal processing (?)







Graph inference

Introduction into Graph Inference

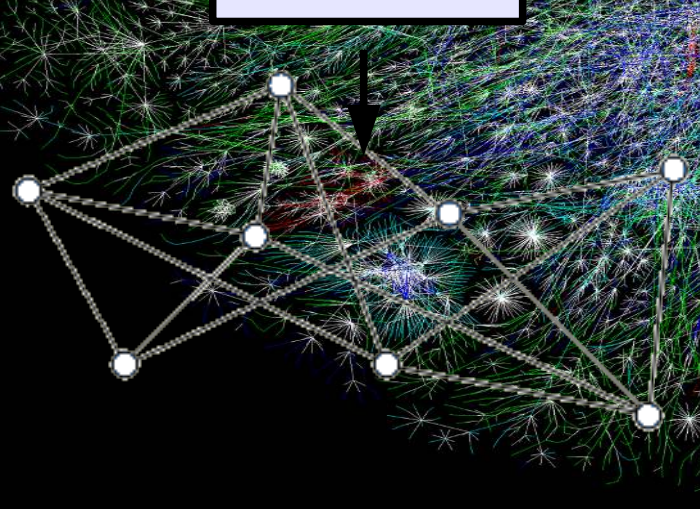
Signals:

Need for a model

Graph
inference

Graphs

Signals and data



Graph inference

Introduction into Graph Inference

Signals:

Need for a model

► Stationary processes

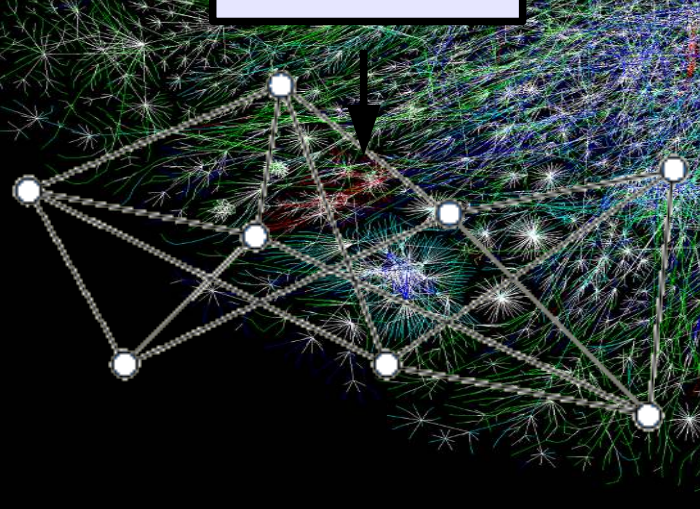
Graph
inference

Graphs:

Need for a model

► Valid for stationary processes

► Desired features



Graph filters in a nutshell

Introduction: motivation and scope

Definition

A stochastic graph signal $\mathbf{x}$ defined on the vertices of a graph G is called *graph wide-sense (or second order) stationary (GWSS)*, if and only if it satisfies the following properties:

Its first moment is constant over the vertex set: $m_{\mathbf{x}}[i] = \mathbb{E}\{\mathbf{x}[i]\} = c \in \mathbb{R}$

Its covariance is invariant with respect to the localization operator

$$\gamma_{\mathbf{x}}[i, n] = \mathbb{E}\{(\mathbf{x}[i] - m_{\mathbf{x}})(\mathbf{x}[n] - m_{\mathbf{x}})\} = T_i \gamma_{\mathbf{x}}[n]$$

Example

White noise is GWSS

If a graph filter g is applied to a GWSS signal, the result remains GWSS

Chapter 11

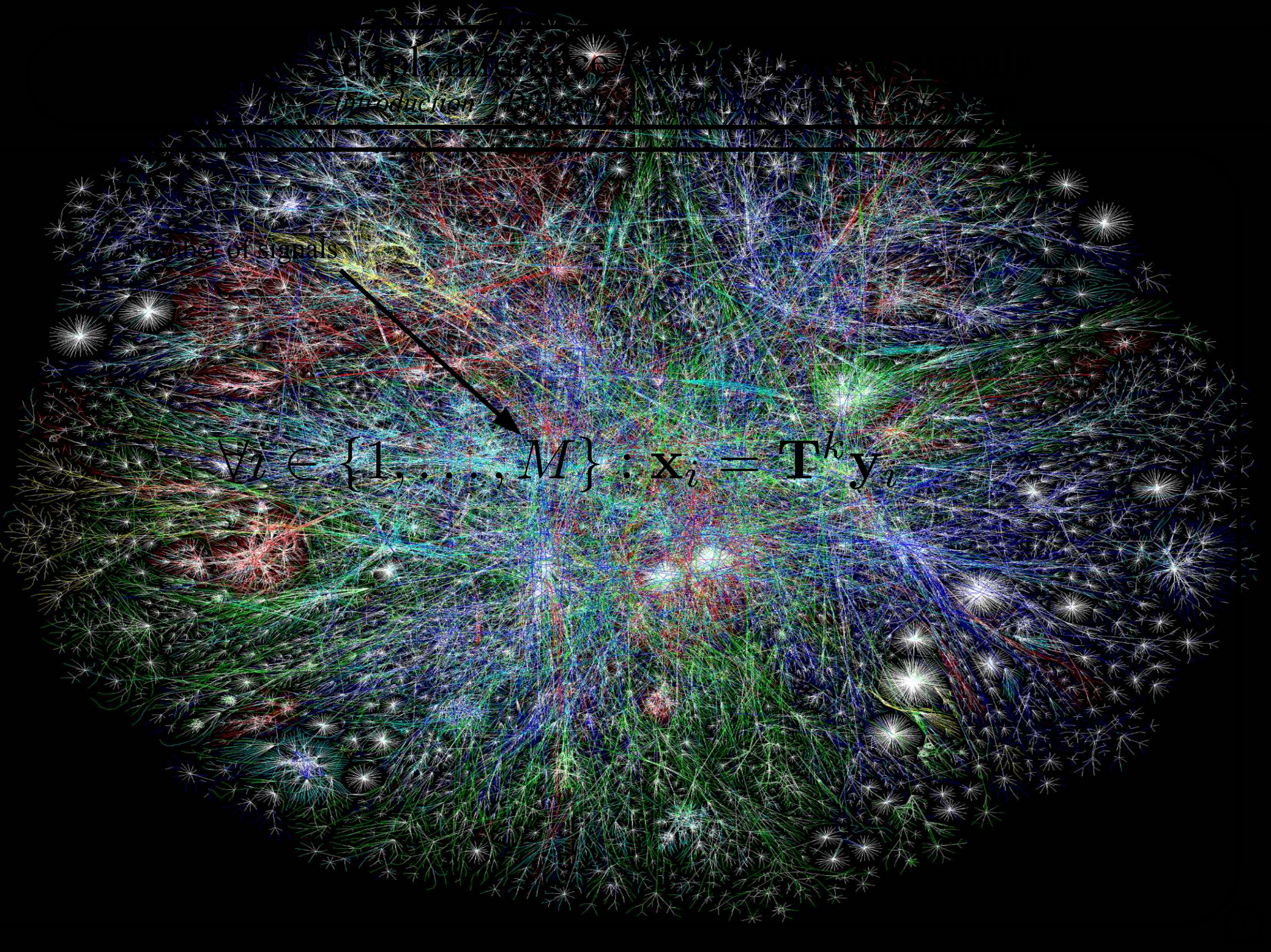
Introduction

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph mining

Introduction Algorithms Applications

Number of signals

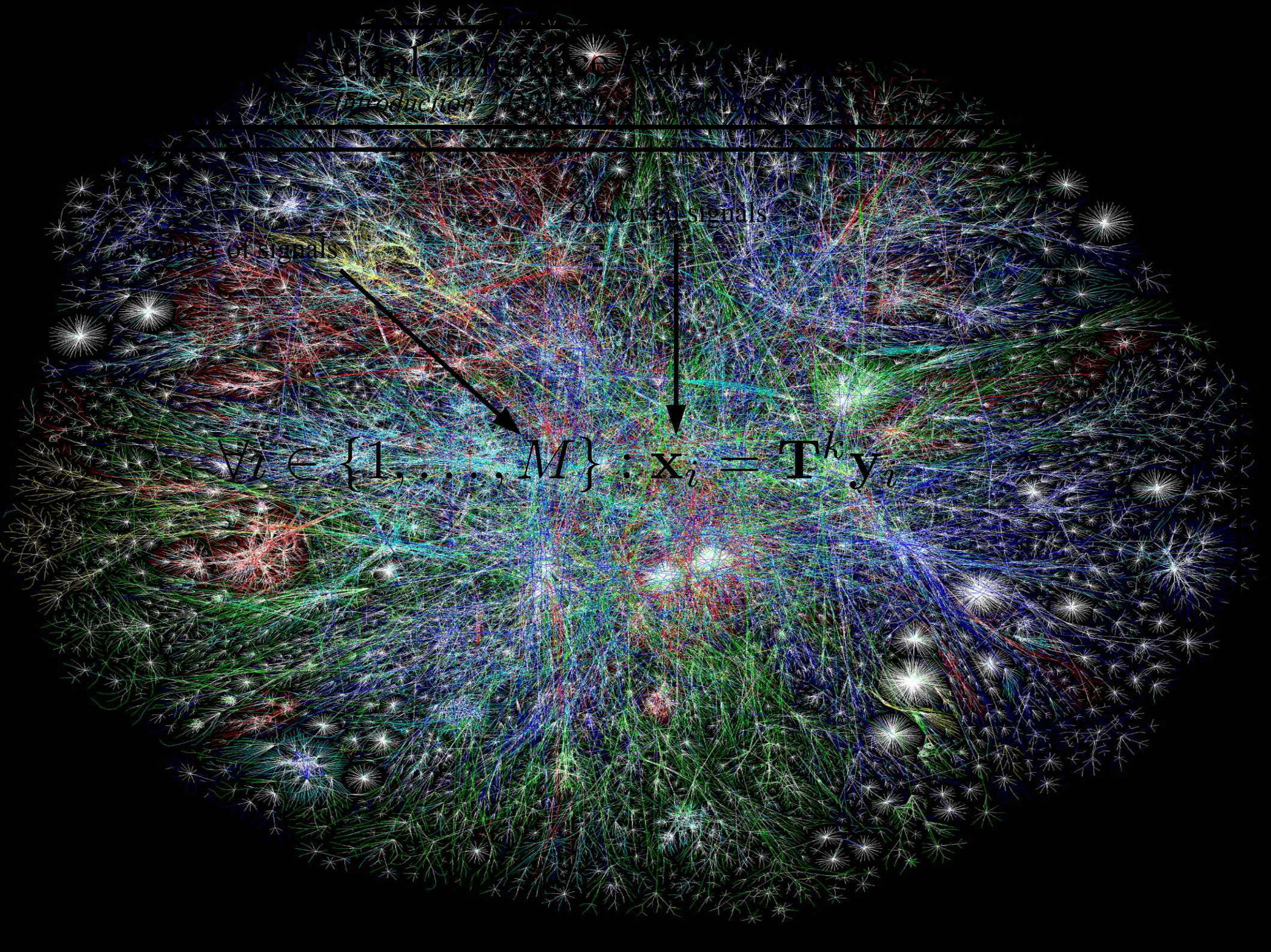

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph mining

Introduction Algorithms Applications

Observed signals

Number of signals


$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph mining in a dynamic network

Introduction Overview Generalization Application

Observed signals

Number of signals

$1/d$ signals

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph diffusion on a graph

Introduction Diffusion Generalization

Observed signals

Number of signals

$1 + d$ signals

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph shift operator used
for diffusion

Graph diffusion on a path

Introduction Definition Generalization

Observed signals

Number of signals

$1 + d$ signals

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph shift operator used
for diffusion

Variable number of diffusion
clinical combination levels

Graph diffusion on a path

Introduction Diffusion Generalization

Observed signals

Number of signals

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$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph shift operator used
for diffusion

Variable number of diffusion
diffusion combination levels

Graph Laplacian

University of Cambridge, 2017

the focus

diffusion context

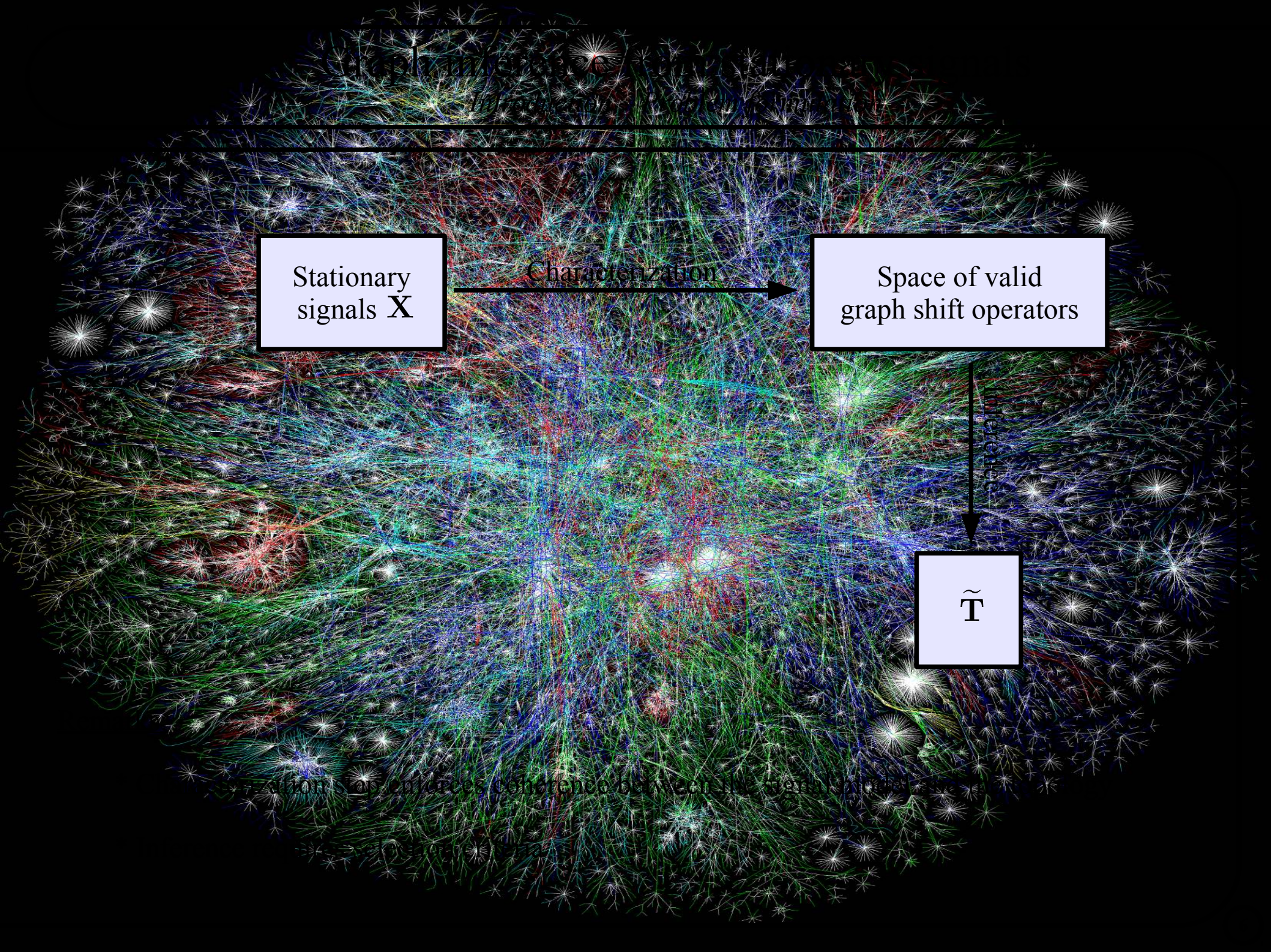
$\mathbf{T}$ is the heat propagation operator $\mathbf{L} = \mathbf{I}_N - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2} = \mathbf{I}_N - \mathbf{T}$

Some properties of $\mathbf{T}$

Eigenvalues in $[-1, 1]$ with at least one equal to 1 (connected graph)

Empty diagonal and positive off-diagonal entries

more general case of stationary processes can be found



Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Interference

$\tilde{\mathbf{T}}$

Stationary Lap enforces coherence between the agents' signals

► "Recent" work on graph inference from stationary signals

- * Introduction / problem formulation
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- * Characterization of the space of admissible matrices
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► Ongoing and future work

- * Application of graph inference to signals classification
- * Tropical graph signal processing (?)

In this paper, we consider the threshold estimator and number of convergence norm. In order to be suitable for the adaptive of the estimator, we show that the threshold estimator from a sample paper which

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Submitted to the *Annals of Statistics*

BY T. TONY CAI* AND HARRISON H. ZHOU†
University of Pennsylvania and Yale University

This paper considers estimation of sparse covariance matrices and establishes the optimal rate of convergence under a range of matrix operator norm and Bregman divergence losses. A major focus is on the derivation of a rate sharp minimax lower bound. The problem exhibits new features that are significantly different from those that occur in the conventional nonparametric function estimation problems. Standard techniques fail to yield good results and new tools are often needed.

We first develop a lower bound technique that is particularly well suited for treating "two-directional" problems such as estimating sparse covariance matrices. The result can be viewed as a generalization of Le Cam's method in one direction and Assouad's Lemma in another. This lower bound technique is of independent interest and can be used for other matrix estimation problems.

We then establish a rate sharp minimax lower bound for estimating sparse covariance matrices under the spectral norm by applying the general lower bound technique. A thresholding estimator is shown to attain the optimal rate of convergence under the spectral norm. The results are then extended to the general matrix ℓ_w operator norms for $1 \leq w \leq \infty$. In addition, we give a unified result on the minimax rate of convergence for sparse covariance matrix estimation under a class of Bregman divergence losses.

1. Introduction. Minimax risk is one of the most widely used benchmarks for optimality and substantial efforts have been made on developing minimax theories in the statistics literature. A key step in establishing a minimax theory is the derivation of minimax lower bounds and several effective lower bound arguments based on hypothesis testing have been introduced in the literature. Well known techniques include Le Cam's method, Assouad's Lemma and Fano's Lemma. See Le Cam (1986) and Tsybakov (2009) for

⁶The research of Tony Cai was supported in part by NSF FRG Grant DMS-0854973.

[†]The research of Harrison Zhou was supported in part by NSF Career Award DMS-0645676 and NSF FRG Grant DMS-0854975.

AMS 2000 subject classifications: Primary 62H12; secondary 62F12, 62G09
Keywords and phrases: Assouad's Lemma, Bregman divergence, covariance matrix estimation, Frobenius norm, Le Cam's method, minimax lower bound, spectral norm, optimal rate of convergence, thresholding.

Statistica Sinica

Abstract
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1. Introduction

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The Annals of Statistics
2012, Vol. 40, No. 1, 466–493
DOI: 10.1214/11-AOS967
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BY HAN XIAO AND WEI BIAO WU
University of Chicago

We obtain a sharp convergence rate for banded covariance matrix estimators of stationary processes. A precise order of magnitude is derived for the spectral radius of sample covariance matrices. We also consider a thresholded covariance matrix estimator that can better characterize sparsity if the true covariance matrix is sparse. As our main tool, we implement Toeplitz [Math. Ann. 70 (1911) 351–376] idea and relate eigenvalues of covariance matrices to the spectral densities or Fourier transforms of the covariances. We develop a large deviation result for quadratic forms of stationary processes using m -dependence approximation, under the framework of causal representation and physical dependence measures.

1. Introduction. One hundred years ago, in 1911, Toeplitz obtained a deep result on eigenvalues of infinite matrices of the form $\Sigma_\infty = (a_{s-t})_{s,t=-\infty}^\infty$. We say that λ is an eigenvalue of Σ_∞ if the matrix $\Sigma_\infty - \lambda \text{Id}_\infty$ does not have a bounded inverse, where Id_∞ denotes the infinite-dimensional identity matrix. Toeplitz proved that, interestingly, the set of eigenvalues is the same as the image set $\{g(\theta), \theta \in [0, 2\pi]\}$, where

$$(1) \quad g(\theta) = \sum_{i \in \mathbb{Z}} a_i e^{i i \theta} \quad \text{with } i = \sqrt{-1}.$$

Note that $g(\theta)$ is the Fourier transform of the sequence $(a_i)_{i=-\infty}^{\infty}$. For a finite $T \times T$ matrix $\Sigma_T = (a_{i-j})_{1 \leq i, j \leq T}$, its eigenvalues are approximately equally distributed (in the sense of Hermann Weyl) as $\{g(\omega_s), s = 0, \dots, T-1\}$, where $\omega_s = 2\pi s/T$ are the Fourier frequencies. See the excellent monograph by Grenander and Szegő (1958) for a detailed account.

Covariance matrix is of fundamental importance in many aspects of statistics including multivariate analysis, principal component analysis, linear discriminant analysis and graphical modeling. One can infer dependence structures among variables by estimating the associated covariance matrices. In the context of stationary time series analysis, due to stationarity, the covariance matrix is Toeplitz in that,

Received May 2011; revised December 2011.

¹Supported in part by NSF Grants DMS-09-06973 and DMS-11-06970.

MSC2010 subject classifications. Primary 62M10; secondary 62H12.

Key words and phrases. Autocovariance matrix, banding, large deviation, physical dependence measure, short range dependence, spectral density, stationary process, tapering, thresholding, Toeplitz matrix.

$$\lambda \|\Theta\|_1 + \log \det(\Theta) + \lambda \|\Theta\|_1$$

Sparse inverse covariance estimation with the graphical lasso

JEROME FRIEDMAN^{*}
TREVOR HASTIE[†]
and ROBERT Tibshirani[‡]

November 17, 2007

Abstract

We consider the problem of estimating sparse graphs by a lasso penalty applied to the inverse covariance matrix. Using a coordinate descent procedure for the lasso, we develop a simple algorithm—the *Graphical Lasso*—that is remarkably fast: it solves a 1000 node problem ($\sim 500,000$ parameters) in at most a minute, and is 30 to 4000 times faster than competing methods. It also provides a conceptual link between the exact problem and the approximation suggested by Meinshausen & Bühlmann (2006). We illustrate the method on some cell-signaling data from proteomics.

1 Introduction

In recent years a number of authors have proposed the estimation of sparse undirected graphical models through the use of L_1 (lasso) regularization. The basic model for continuous data assumes that the observations have a multivariate Gaussian distribution with mean μ and covariance matrix Σ . If the i th component of Σ^{-1} is zero, then variables i and j are conditionally

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GENERALIZED

Department of

Graph signal processing on the vertices of a graph, pretation of the eigenvectors transform for graph signals as a precision matrix of signals, and we propose a tion takes a covariance matrix one row/column of the quadratic program. Exported Laplacian precision matrix nonzero entries and it can graphical Lasso. For text, follow the orientation. We ones obtained using other

Index Terms— graph signal processing, Gaussian Markov random fields (GMRFs).

Graph signal processing on high dimensional data vertices of a weighted graph techniques by interpreting graph frequencies and the (GFT). Graph structures of sensor networks [1], brain image and video coding fields is that of learning the graph must have a meaning. Also, the learning algorithm dimensions increase.

In this work, we formulate problem and focus on learning a graph from data can V and a set of corresponding E and edge weights, or nonzero patterns define the pose a method that takes a set of entries, for example an puts a generalized Laplacian definite matrix with n Our solution Q satisfies: abilitic interpretation, ii) past data representation, iii) Using GLs instead of has several advantages: i)

Our solution Q satisfies: abilitic interpretation, ii) past data representation, iii) Using GLs instead of has several advantages: i)

This work was supported

978-1-4799-9988-0/16

Graph Learning from Data under Structural and Laplacian Constraints

Hilmi E. Egilmez, *Student Member, IEEE*, Eduardo Pavez, *Student Member, IEEE*, and Antonio Ortega, *Fellow, IEEE*

Abstract—Graphs are fundamental mathematical structures used in various fields to represent data, signals and processes. In this paper, we propose a novel framework for learning/estimating graphs from data. The proposed framework includes (i) formulation of various graph learning problems, (ii) their probabilistic interpretations and (iii) efficient algorithms to solve them. We specifically focus on graph learning problems where the goal is to estimate a graph Laplacian matrix from some observed data under given structural constraints (e.g., graph connectivity and sparsity). Our experimental results demonstrate that the proposed algorithms outperform the current state-of-the-art methods in terms of graph learning performance.

Index Terms—Graph learning, sparse graph learning, graph estimation, optimization, graph Laplacian matrices, Gaussian Markov random fields (GMRFs).

1. INTRODUCTION

GRAPHS are generic mathematical structures consisting of sets of vertices and edges, which are used for modeling pairwise relations (edges) between objects (vertices). In practice, this representation is often extended to weighted graphs, for which a set of scalar values (i.e., weights) are assigned to edges and potentially vertices. Thus, weighted graphs offer general and flexible representations to model affinity relations between the objects of interest.

Many practical problems can be represented using weighted graphs, including for example a large class of combinatorial problems such as matching, shortest-path and network flow [1]. In signal/data-oriented problems, weighted graphs can provide simplified (e.g., sparse) representations for robust modeling of signal/data [2], which are also useful for analyzing and visualizing the relations between samples/features. Moreover, weighted graphs naturally emerge in networked data applications, such as learning, data processing and analysis on computer, social, sensor, energy, transportation and biological networks [3], where signal/data samples are inherently structured on a graph associated with the underlying network. Similarly, image/video signals are often considered to be structured on a weighted graph, where the weights capture the correlation between neighboring pixel values [4]–[6].

In this paper, we focus on signal/data-oriented problems, where our goal is learning a graph (i.e., graph-based model) to characterize the relationship between the entries of a signal/data vector from multiple observations. In addition to signal/data observations, if the graph structure (graph topology) is

known, then the graph learning problem reduces to estimation of graph weights. On the other hand, if it is partially known or unknown, then graph learning involves estimation of both graph weights and the graph structure. Also, learning sparse graphs is generally useful for data analysis and algorithmic purposes. In order to address these scenarios, we propose a general framework where graph learning is formulated as the estimation of different types of graph Laplacian matrices from data, which lead to weighted graphs with nonnegative edge weights (see Definitions 1 and 2 in Section II). Specifically, our formulations involve, for a given data statistic S (e.g., empirically obtained covariance/correlation matrices), minimization of objective functions of the following form:

$$\underbrace{\mathcal{R}(\Theta S)}_{\mathcal{R}(\Theta S)} - \log \det(\Theta) + \underbrace{(n/2)\|\Theta\|_{\text{sp}}}_{\mathcal{R}(\Theta)} \quad (1)$$

where Θ is the target variable matrix (i.e., a specific type of graph Laplacian), $\mathcal{R}(\Theta, n)$ denotes the sparsity promoting regularization term with parameter n [9], [10], and $\mathcal{D}(\Theta, S)$ is the data fidelity term, whose minimization is equivalent to the maximum likelihood estimation of inverse covariance (or precision) matrices under certain probabilistic assumptions¹. In addition to the objective in (1), we consider problem-specific structural and Laplacian constraints based on (i) available structural information about the graph, and (ii) the desired type of Laplacian. In this work, we focus on three types of graph Laplacians, namely, *generalized graph Laplacians*, *diagonally dominant generalized graph Laplacians*, and *combinatorial graph Laplacians*, and develop novel techniques to estimate them from data (i.e., data statistic S). As illustrated in Fig. 1 and further discussed in Section IV, our proposed graph Laplacian estimation techniques can also be viewed as methods to learn different types of Gaussian-Markov random field (GMRF) models, defined by their mean vectors and precision matrices [11], [12]. The main contributions of this work are summarized as follows:

- **Problem Formulations:** We formulate graph learning problems addressing the estimation of three types of graph Laplacian matrices with structural constraints, which extend prior formulations [13]–[17] considering estimation of two types of Laplacians without structural constraints.
- **Probabilistic Interpretations:** We present probabilistic interpretations of our proposed problems by showing that their solutions lead to optimal parameter estimation for special classes of GMRFs, as illustrated in Fig. 1. While

¹Motivation for using this metric is discussed further in later sections.

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Learning Graph Laplacian Matrix in Smooth Graph Signal

Learning Laplacian Matrix in Smooth Graph Signal

Xia

Abstract—The construction of a graph Laplacian matrix is a crucial role in the success of many algorithms for handling emerging field of graph signal processing. However, the graph Laplacian matrix is not always available in the real world. In this paper, we define a smooth graph signal as a signal whose value at each node is defined by the smoothness on the graph. We propose a factor analysis model to learn the graph Laplacian matrix from smooth signals. We impose a Gaussian prior on the graph Laplacian matrix to control these signals. We propose an efficient representation of the graph signals. We propose a framework to learn the graph Laplacian matrix from both synthetic and real world graph signals. We propose a graph learning framework to learn the graph Laplacian matrix from signal observations.

Index Terms—Laplacian matrix, graph signal processing, representation learning

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MODERN data processing and analysis algorithms for handling emerging field of graph signal processing. However, the graph Laplacian matrix is not always available in the real world. In this paper, we define a smooth graph signal as a signal whose value at each node is defined by the smoothness on the graph. We propose a factor analysis model to learn the graph Laplacian matrix from smooth signals. We impose a Gaussian prior on the graph Laplacian matrix to control these signals. We propose an efficient representation of the graph signals. We propose a framework to learn the graph Laplacian matrix from both synthetic and real world graph signals. We propose a graph learning framework to learn the graph Laplacian matrix from signal observations.

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D. Thum, F. Fomont and S. K. Saha are with the Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland (e-mail: {thum, fomont, sasha}@epfl.ch).
X. Deng is supported by a research grant from the Swiss National Science Foundation (SNSF) under the project LEXAN, and the SNSF project 15490.

How to learn a graph from smooth signals

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Signal Processing Laboratory 2 (LTS2)
Swiss Federal Institute of Technology Lausanne (EPFL), Switzerland

Abstract

We propose a framework that learns the graph structure underlying a set of smooth signals. Given $X \in \mathbb{R}^{m \times n}$ whose rows reside on the vertices of an unknown graph, we learn the edge weights $w \in \mathbb{R}^{m(m-1)/2}$ under the smoothness assumption that $\text{tr}(X^T L X)$ is small. We show that the problem is a weighted ℓ_1 minimization that leads to naturally sparse solutions. We point out how known graph learning or construction techniques fall within our framework and propose a new model that performs better than the state of the art in many settings. We present efficient, scalable primal-dual based algorithms for both our model and the previous state of the art, and evaluate their performance on artificial and real data.

1 INTRODUCTION

We consider a matrix $X \in \mathbb{R}^{m \times n} = [x_1, \dots, x_n]^T$, where each row $x_i \in \mathbb{R}^n$ resides on one of m nodes of a graph G . Then each of the n columns of X can be seen as a signal on the same graph. A simple assumption about data residing on graphs, but also the most widely used one is that it changes smoothly between connected nodes. An easy way to quantify how smooth is a set of vectors $x_1, \dots, x_n \in \mathbb{R}^n$ on a given weighted undirected graph is through the function

$$\frac{1}{2} \sum_{i,j} W_{ij} \|x_i - x_j\|^2,$$

where W_{ij} denotes the weight of the edge between nodes i and j . In words, if two vectors x_i and x_j from a smooth set reside on two well connected nodes (so

W_{ij} is big), they are expected to have a small distance. Using the graph Laplacian matrix $L = D - W$, where D is the diagonal degree matrix with $D_{ii} = \sum_j W_{ij}$, this function can be written in matrix form as

$$\text{tr}(X^T L X).$$

The importance of the graph Laplacian has long been known as a tool for embedding, manifold learning, clustering and semisupervised learning, see e.g. Belkin and Niyogi (2001), Belkin, Niyogi, and Sindhwani (2006), and Zhu, Chakranarayanan, Lafferty, et al. (2003). More recently we find an abundance of methods that exploit this notion of smoothness to regularize various machine learning tasks, solving problems of the form

$$\underset{X}{\text{minimize}} \quad g(X) + \text{tr}(X^T L X), \quad (1)$$

Zhang, Popescul, and Dom (2006) use it to enhance web page categorization with graph information, Zheng et al. (2011) for graph regularized sparse coding, Cai et al. (2011) use the same term to regularize NMF, Jiang et al. (2013) for PCA and Katsifolios et al. (2014) for matrix completion. Having good quality graphs is key to the success of the above methods.

The goal of this paper is to solve the complementary problem of learning a good graph:

$$\underset{L \in \mathcal{L}}{\text{minimize}} \quad \text{tr}(X^T L X) + f(L), \quad (2)$$

where $\mathcal{L}$ denotes the set of valid graph Laplacians.

Why is this problem important? Firstly because it enables us to directly learn the hidden graph structure behind our data. Secondly because in most problems that can be written in the form of eq. (1), we are often given a noisy graph, or no graph at all. Therefore, starting from the initial graph and alternating between solving problems (1) and (2) we can at the same time get a better quality graph and solve the task of the initial problem.

Related Work. Dempster (1972) was one of the first to propose the problem of finding connectivity from

$$S(\mathbf{x}) = \sum_{\{u,v\} \in \mathcal{E}} \mathbf{W}[u,v] (\mathbf{x}[u] - \mathbf{x}[v])^2$$

$$\min_{L, \mathbf{y}} \|\mathbf{x} - \mathbf{y}\|_2^2 + \alpha \mathbf{y}^\top \mathbf{L} \mathbf{y}$$

Interplay between Diffusion and Graph Learning

► Close to a valid solution?

Graph reconstruction from the observation of diffused signals*

Bastien Padeloup¹, Michael Rabut², Vincent Gripon¹, Dominique Pastor¹ and Grégoire Mercier³

Abstract—Signal processing on graphs has received a lot of attention in the recent years. A lot of techniques have arisen, inspired by classical signal processing ones, to allow studying signals on any kind of graph. A common aspect of these techniques is that they require a graph correctly modeling the studied support to explain the signals that are observed on it. However, in many cases, such a graph is unavailable or has no real physical evidence. An example of this latter case is a set of sensors randomly thrown in a field which obviously observe related information. To study such signals, there is no intuitive choice for a support graph. In this document, we address the problem of inferring a graph structure from the observation of signals, under the assumption that they were issued from the diffusion of initially *i.i.d.* signals. To validate our approach, we design an experimental protocol, in which we diffuse signals on a known graph. Then, we forget the graph, and show that we are able to retrieve it very precisely from the only knowledge of the diffused signals.

1. INTRODUCTION

Signal processing on graphs has recently gained a lot of interest, and recent years have seen the appearance of numerous new techniques based on its principles. Although it was originally seen as an extension of classical signal processing to more complex domains – thus generalizing the Fourier approach –, applications now emerge in various fields such as wavelets on graphs, that find application in fields such as brain imagery [12]; or filtering, applied to big data analysis [16]. Other examples are the recent work of Dong *et al.* [7] that have studied the political orientations of the Swiss cantons from voting data; or the application of Ahly *et al.* [1] to photographic prints classification.

One common aspect of all the existing applications is that they rely on the Graph Fourier Transform (GFT) of the graphs, which is a cornerstone of signal processing on graphs. This particular transform is performed using the eigenvectors of a particular matrix that is dependent on the graph used to represent the support of the signals. However, such a matrix is not always available. In some even more complex – and frequent – cases, the graph has no real physical evidence. Example of this are when sensors are thrown in a field, or are used to measure the brain activity (EEG). There is an

obvious link between the signal components observed in the various signals, but there is no intuitive notion of graph. To circumvent this problem, methods have recently been developed to allow one to recover a graph structure from the observation of signals. The idea behind this is to provide a structure that would *naturally* represent the relationships between the signal components. Even in the case when the graph does not exist, such techniques should be able to provide a graph that captures these relationships, and would provide a support for the application of graph signal processing tools.

One possible approach was proposed in the Ph.D. dissertation of Costard [5]. Her work is based on previous work from Dempster [6], that estimates the covariance matrix of signals with the constraint that some values of its inverse – called the precision matrix – are null. This latter work has then allowed Wernisch [19] to show that the non-null entries of this precision matrix correspond to edges of a graph representing the studied signals. Based on this result and on the additional works in [15], [2] on model selection, Costard [5] reviews existing methods for recovering a sparse precision matrix. Additionally, she introduces a new method called ABIGlasso, based on Bayesian inference techniques initialized using the Graphical lasso [9]. Related practical algorithms for estimating a sparse inverse covariance matrix were developed since then (eg. [10]).

Another existing approach to retrieve a graph from the observation of signals is the one introduced in the recent work of Dong *et al.* [7]. The notion of smoothness of a signal on a graph was originally defined in [17] as a quantity representing the total variation of a signal with respect to adjacent nodes. Based on the assumption that adjacent nodes in a graph should share similar signal values – i.e. signals should be smooth – Dong *et al.* propose a graph reconstruction technique that enforces this smoothness constraint.

To the best of our knowledge, these two approaches are the main efforts that have been made to answer the problem of graph reconstruction from the observation of signals. However, for the approach of Costard, the author shows that her work is only applicable to a small number of variables [5]. Concerning the approach of Dong *et al.*, they use the assumption that signals should be smooth on the recovered graph. In this paper, we propose a new method to recover a graph from signals, with different assumptions on the observed signals; that is to say they are issued from the diffusion of initially *i.i.d.* signals in the graph. Our method consists of searching a solution in a system of linear constraints, which is of polynomial complexity [3] and can

*This work was supported by the European Research Council under the European Union's Seventh Framework Programme (FP7/2007-2013) / ERC grant agreement n° 290901. Additional thanks go to the CNRS of Université de Bretagne and to the IIS research group for the financial support.

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Network Topology Inference from Spectral Templates

Samir

Abstract—We address the problem of inferring the network topology from the observation of signals. The fresh look at convex optimization and the identification of the graph shift operator (GSO) given only its eigenvalues, e.g., from the signals diffused on the graph shift that, while less information, endows the network structure with certain desired properties such as sparsity. The focus is on developing efficient recovery algorithms along with identifiability conditions for two particular shifts, the adjacency matrix and the normalized graph Laplacian. Application domains include network topology identification from steady-state signals generated by a diffusion process, and design of a graph filter that facilitates the distributed implementation of a prescribed linear network operator. Numerical tests showcase the effectiveness of the proposed algorithms in recovering synthetic and structural brain networks.

Index Terms—Network topology inference, graph signal processing, spectral graph theory, principal component analysis

1. INTRODUCTION

Advancing a holistic theory of networks necessitates fundamental breakthroughs in modeling, identification, and controllability of distributed network processes – often conceptualized as *signals defined on the vertices* of a graph [1, 2]. Under the assumption that the signal properties are related to the topology of the graph where they are supported, the goal of graph signal processing (GSP) is to develop algorithms that fruitfully leverage this relational structure [3, 4]. Instrumental to that end is the so-called graph-shift operator (GSO) [4], a matrix capturing the graph's local topology and whose eigenvectors are central to defining graph Fourier transforms [5]. Most GSP works assume that the GSO (hence the graph) is known, and then analyze how the algebraic and spectral characteristics of the GSO impact the properties of the signals and filters defined on such a graph. Here instead we take the reverse path and investigate how to use information available from graph signals and filters to infer the underlying graph topology; see also [6, 7]. In a nutshell, we follow a two-step approach whereby we first leverage results from GSP theory to identify the GSO's eigenspace using the available information, and then rely on these spectral templates to recover the GSO itself.

Network topology inference from a set of (graph-signal) observations is a prominent problem in Network Science [2, 8]. Since

NETWORK TOPOLOGY IDENTIFICATION FROM SPECTRAL TEMPLATES

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ABSTRACT

Network topology inference is a cornerstone problem in statistical analyses of complex systems. In this context, the fresh look advocated here permeates benefits from convex optimization and graph signal processing, to identify the so-called graph shift operator (referred to the network topology) given only the eigenvectors of the shift. These spectral templates can be obtained, for example, from principal component analysis of a set of graph signals defined on the particular network. The novel idea is to find a graph shift that while being consistent with the provided spectral information, it endows the network structure with certain desired properties such as sparsity. The focus is on developing efficient recovery algorithms along with identifiability conditions for two particular shifts, the adjacency matrix and the normalized graph Laplacian. Application domains include network topology identification from steady-state signals generated by a diffusion process, and design of a graph filter that facilitates the distributed implementation of a prescribed linear network operator. Numerical tests showcase the effectiveness of the proposed algorithms in recovering synthetic and structural brain networks.

Index Terms—Network topology inference, graph signal processing, spectral graph theory, principal component analysis

1. INTRODUCTION

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Work in this paper is supported by the NSF award CCF-1217960 and Spanish MINECO TEC2013-341084-R.

networks encode similarities between nodes, most approaches construct graphs whose edge weights correspond to the correlation, or coherence, between signal profiles at incident nodes. This approach is not without merit and widely used in practice, but it exhibits several drawbacks, the main one being that links are formed taking into account only pairwise interactions, ignoring that the observed correlations can be due to latent network effects. Acknowledging these limitations, alternative methods rely on partial correlations [2, 9], Gaussian graphical models [10, 11], or Granger causality [8, 12]. Differently, recent GSP-based network inference frameworks postulate that the network exists as a latent underlying structure, and that observations are generated as a result of a network process defined in such a graph. For instance, network structure is estimated in [7] to unveil unknown relations among node time series adhering to an autoregressive model involving graph-filter dynamics. A factor analysis-based approach was put forth in [8] to infer graph Laplacians, seeking that input graph signals are smooth over the learned topologies. Different from [6–7] that operate on the graph domain, the goal here is to identify graphs that endow the given observations with desired spectral (frequency-domain) characteristics.

After surveying the required GSP background in Section 2, we formulate the GSO identification problem given spectral templates (Section 3). The novel idea is to search among all feasible networks for the one that endows the resulting graph-signal transforms with prescribed spectral properties, while the inferred graph also exhibits desirable structural characteristics. Optimization problems are formulated to recover two particular GSOs, namely the adjacency matrix (Section 3.1) and the normalized graph Laplacian (Section 3.2). Conditions under which the feasible set reduces to a singleton are derived, and convex relaxations leading to computationally-efficient algorithms are proposed for the general case. Section 4 outlines two pragmatic scenarios where acquisition of the GSO eigenspace is well motivated, making the case that setups where the GSO is unknown but its spectral templates are available can be more the rule than the exception. Computer simulations highlight the effectiveness of the proposed algorithms in identifying both synthetic and real-world networks (Section 5). Concluding remarks are given in Section 6.

2. GRAPH SIGNALS AND GRAPH FILTERS

Here we formally introduce notation and terminology as well as different GSP tools that will be used throughout the paper. Graphs. Let $\mathcal{G}$ denote an undirected graph with a set of nodes $\mathcal{V}$ with cardinality N and a set of links $\mathcal{E}$, such that if nodes i and j are connected to i , then both (i, j) and (j, i) belong to $\mathcal{E}$. The set $\mathcal{N}_i = \{j \mid (i, j) \in \mathcal{E}\}$ stands for the neighborhood of i . For any given $\mathcal{G}$ we define the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ as a sparse matrix with non-zero elements $A_{ij} = A_{ji}$ if and only if $(i, j) \in \mathcal{E}$. The values of A_{ij} can be binary, or real in the weighted case to

This paper presents a convex optimization approach for estimating the graph structure and adjacency matrix. The algorithm is demonstrated on time series datasets, and is compared to that of the existing methods. The proposed method is shown to be able to handle data and consistent real datasets.

Index Terms— Graph, Adjacency Matrix.

I. INTRODUCTION

Networked data are everywhere in geology and biology. While there are many different types of networks, they are growing rapidly. While there are many different types of networks, they are growing rapidly. While there are many different types of networks, they are growing rapidly.

In this paper, we consider data in the form of a matrix A . Current work largely associates graph theory of the processes is instead associated with the graphing inspiration from the (DSP) framework [3]. We first provide a brief review of relevant concepts, introduce our new network reconstruction algorithm to be generated by this process simulation results in section 2.

This work is partially supported by the Swiss National Science Foundation (SNF).

978-1-4673-6997-8/15/\$16.00 © 2015 IEEE

arXiv:1611.01456v1 [cs.LG] 4 Nov 2016

Abstract—Effective inference for identifying the structure of a network is often represented by a graph. In particular, the global structure of a network is not readily available. In particular, the global structure of a network is not readily available. In particular, the global structure of a network is not readily available.

Index Terms—Laplacian, representation theory.

1. **INTRODUCTION**
Data analysis and processing of structured data is one of the most important tasks in many domains, such as computer networks, brain vision. Graphs are common in many domains, such as computer networks, brain vision. Graphs are common in many domains, such as computer networks, brain vision.

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S. Ding is with the MIT Media Lab.

D. Krieger is with the Performance Computing (ASC) (EPFL), Lausanne, Switzerland.

Learning heat diffusion graphs

GRAPH LEARNING UNDER SPARSITY PRIORS

Hermine Petric Maretic, Dorothea Thanou and Pascal Frossard

Signal Processing Laboratory (LTS4), EPFL, Switzerland

ABSTRACT

Graph signals offer a very generic and natural representation for data that lives on networks or irregular structures. The actual data structure is however often unknown a priori but can sometimes be estimated from the knowledge of the application domain. If this is not possible, the data structure has to be inferred from the mere signal observations. This is exactly the problem that we address in this paper, under the assumption that the graph signals can be represented as a sparse linear combination of a few atoms of a structured graph dictionary. The dictionary is constructed on polynomials of the graph Laplacian, which can sparsely represent a general class of graph signals composed of localized patterns on the graph. We formulate a graph learning problem, whose solution provides an ideal fit between the signal observations and the sparse graph signal model. As the problem is non-convex, we propose to solve it by alternating between a signal sparse coding and a graph update step. We provide experimental results that outline the good graph recovery performance of our method, which generally compares favourably to other recent network inference algorithms.

Index Terms— graph learning, graph signal processing, Laplacian matrix, sparse signal prior, graph dictionary

I. INTRODUCTION

Graphs provide a flexible tool for modelling and manipulating complex data that resides on topologically complicated domains such as transportation networks, social, and computer networks, brain analysis or even digital images. Typically, once a graph is well-defined, classical data analysis and inference tasks can be effectively performed by using tools such as spectral graph theory or generalization of signal processing notions in the graph domain [1]. However, it is often the case that the graph structures are not defined a priori or the intrinsic relationships between different signal observations are not clear or are defined only in a very subtle manner. Since the definition of a meaningful graph plays a crucial role in the analysis and processing of structured data, the inference of the graph itself from the data is a very important problem that has not been well investigated so far.

The definition of the graph has been initially studied in a machine learning framework with simple models such as K-nearest neighbour graphs and other heuristically defined graph kernels [2], [3] or convex combinations of them [4]. Richer adaptivity to the data is obtained by relating

the graph structure more closely to data properties in order to infer a graph topology. Techniques such as sparse inverse covariance estimation [5], [6] rely on Gaussian graphical models to identify partial correlations between random variables and define graph edges. Such techniques have also been extended to very large graphs, by learning highly structured representations [7]. More recent works relax the assumption of a full rank precision matrix by inferring the graph topology from a graph signal processing perspective [8], [9], [10], [11] under explicit graph signal smoothness assumptions. The above works assume that the data globally evolves smoothly on the underlying structure. However, such a model might not be very precise for many real world datasets, which can feature highly localized behaviors or piecewise smoothness. The recent framework in [12] that observes graph signals as white signals filtered with a graph shift operator polynomial, is one of the first network inference works to depart from explicit global smoothness assumptions. A similar idea uses adjacency matrix polynomials to model causation in time-varying signals [13], [14].

In this work, we consider a generic model where the graph signals are represented by (sparse) combinations of overlapping local patterns that reside on the graph. That is, given a set of graph signals, we model these signals as a linear combination of only a few components (i.e., atoms) from a graph structured dictionary that captures localized patterns on the graph. We incorporate the underlying graph structure into the dictionary through the graph Laplacian operator. In order to ensure that the atoms are localized in the graph vertex domain, we further impose the constraint that our dictionary is a concatenation of substructures that are polynomials of the graph Laplacian [15]. Based on this generic model, we cast a new graph learning problem that aims at estimating a graph that explains the data observations, which should eventually form a sparse set of localized patterns on the learned graph. We propose an alternating optimization algorithm to address the resulting non-convex inverse problem, which is based on a sparse coding step optimized with orthogonal matching pursuit (OMP) and a graph learning step performed by applying a projected gradient descent step. We finally provide a few illustrative experiments on synthetic data, and we show that our generic graph signal model leads to better graph recovery performance than state-of-the-art algorithms.

The rest of the paper is organized as follows. We describe our graph learning framework in Section 2. We then present our alternating optimization algorithm in Section 3, and evaluate the graph recovery performance in Section 4.

Reconstruction

Abstract—This paper presents the topology of an unknown network reconstructed by a consensus dynamic system. The network is reconstructed by a consensus dynamic system. The network is reconstructed by a consensus dynamic system.

I. INTRODUCTION

In recent years, considerable interest has been shown in the reconstruction of the network structure from a series of dynamical responses. Moreover, we propose a second algorithm to recover the exact structure of the network (including edge weights), as well as the power spectral density of the input noises, when an eigenvalue-eigenvector pair of the connectivity matrix is known (for example, Laplacian connectivity matrices). Finally, for the particular cases of nonreciprocal networks (i.e., networks with no directed edges pointing in opposite directions) and undirected networks, we propose specialized algorithms that result in a lower computational cost.

arXiv:1303.3250v2 [cs.SI] 15 Mar 2013

arXiv:1308.2248v1 [cs.SI] 9 Aug 2013

Topology Identification of Directed Dynamical Networks via Power Spectral Analysis

Shahin Shahrampour and Victor M. Preciado, *Member, IEEE*,

Abstract

We address the problem of identifying the topology of an unknown weighted, directed network of LTI systems stimulated by wide-sense stationary noises of unknown power spectral densities. We propose several reconstruction algorithms based on the cross-power spectral densities of the network's response to the input noises. Our first algorithm reconstructs the Boolean structure (i.e., existence and directions of links) of a directed network from a series of dynamical responses. Moreover, we propose a second algorithm to recover the exact structure of the network (including edge weights), as well as the power spectral density of the input noises, when an eigenvalue-eigenvector pair of the connectivity matrix is known (for example, Laplacian connectivity matrices). Finally, for the particular cases of nonreciprocal networks (i.e., networks with no directed edges pointing in opposite directions) and undirected networks, we propose specialized algorithms that result in a lower computational cost.

I. INTRODUCTION

The reconstruction of networks of dynamical systems is an important task in many realms of science and engineering, including biology [1]–[4], physics [5]–[7] and finance [8]. In the literature, we find a wide collection of approaches aiming to solve the network reconstruction problem. In the physics literature, we find in [6] a method to identify a network of dynamical systems which assumes that the input of each node can be individually manipulated. In [10], an approach based on Granger's causality [11] and the theory of reproducing kernel Hilbert spaces is proposed. In the statistics community, the reconstruction problem is usually approached using graphical models by associating a random variable to each node and assuming that the (vector-valued) observations are independent and identically distributed. In this setting, Bach and Jordan

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► "Recent" work on graph inference from stationary signals

- * Introduction / problem formulation
- * Related work
- * Characterization of the space of admissible matrices
- * Strategies for selecting a graph in the polytope
- * Numerical experiments
- * Evaluation of inference methods on a real-life dataset

► Ongoing and future work

- * Application of graph inference to signals classification
- * Tropical graph signal processing (?)

Graph inference in a high-dimensional space

Characterization of valid graph shift operators

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

- $\tilde{\mathbf{T}} \triangleq \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \mathcal{X}^*$ is symmetric.
- $\forall i, j \in \{1, \dots, N\}, j \geq i, T(i, j) \geq 0$.
- Let χ_1 be the constant-sign eigenvector in $\mathcal{X}$: $\lambda_1 = 1$.
- $\forall i \in \{1, \dots, N\}: \tilde{\lambda}_i \in [-1, 1]$.

Graph inference in a high-dimensional space

Characterization of stationary signals

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

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$\tilde{\mathbf{T}}$

Eigenvectors of the
covariance matrix

- $\tilde{\mathbf{T}} \triangleq \mathbf{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathbf{X}^*$ is symmetric.
 - $\forall i, j \in \{1, \dots, N\}, j \geq i: \tilde{T}(i, j) \geq 0$.
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 - $\forall i \in \{1, \dots, N\}: \tilde{\lambda}_i \in [-1, 1]$.
- Diffusion matrix properties
- Scale factor

Graph inference in a high-dimensional space

Characterization of valid graph shift operators

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Stationary
signals $\mathbf{X}$

Characterization

Space of valid
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$\tilde{\mathbf{T}}$

Eigenvectors of the
covariance matrix

Unknown

- $\tilde{\mathbf{T}} \triangleq \mathcal{X} \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ 0 & & \lambda_N \end{pmatrix} \mathcal{X}^*$ is symmetric.
 - $\forall i, j \in \{1, \dots, N\}, i \neq j \implies T(i, j) \geq 0$.
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 - $\forall i \in \{1, \dots, N\}: \tilde{\lambda}_i \in [-1, 1]$.
- Diffusion matrix properties
- Scale factor

Graphs

Graph of adjacency matrix

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

$$\tilde{\mathbf{T}} = \mathbf{X} \begin{pmatrix} \widetilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \widetilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$$

$$\tilde{\mathbf{T}}(i, j) = \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N$$

Graphs

Graphs of admittance

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

$$\tilde{\mathbf{T}} = \mathbf{x} \begin{pmatrix} \widetilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \widetilde{\lambda}_N \end{pmatrix} \mathbf{x}^\top$$

$$\tilde{\mathbf{T}}(i, j) = \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N$$

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N \geq 0$$

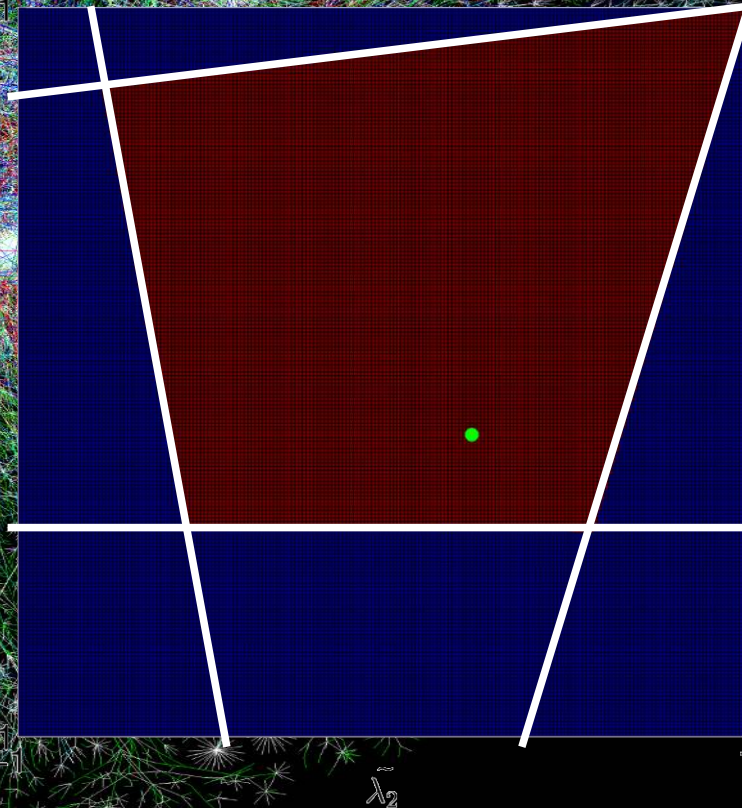
$\pi_1(\mathbb{R}P^3)$ *Characterization of the 3-fold*

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)\lambda_1}_{Q_{231}} + \underbrace{\chi(2,2)\chi(3,2)\lambda_2}_{Q_{232}} + \underbrace{\chi(2,3)\chi(3,3)\lambda_3}_{Q_{233}}$$

Vignettes Characterization of the T(2,3)

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

$$W = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.117 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$



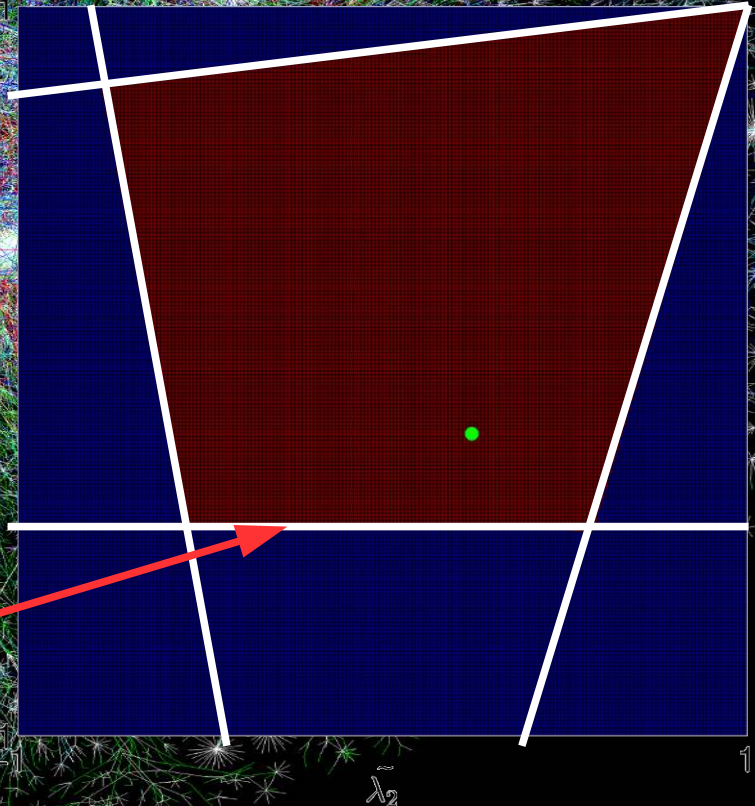
Wigner-Dyson

Characteristic polynomial

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)\lambda_1}_{Q_{1231}} + \underbrace{\chi(2,2)\chi(3,2)\lambda_2}_{Q_{1232}} + \underbrace{\chi(2,3)\chi(3,3)\lambda_3}_{Q_{1233}}$$

$$W = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.117 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$

Constraints



Wigner-Dyson

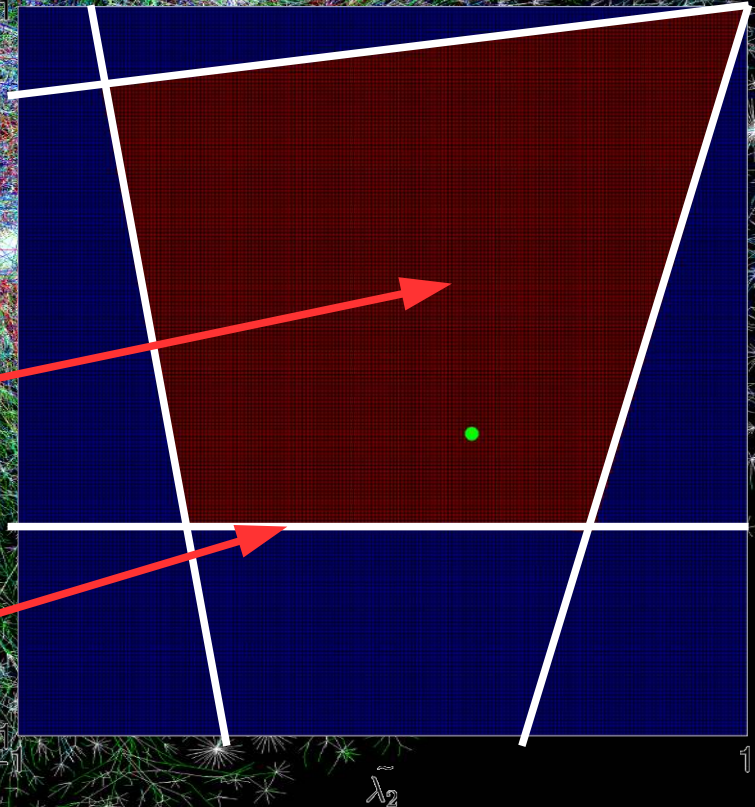
Characteristic polynomial

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}} \lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}} \lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}} \lambda_3$$

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Admissible $\tilde{T}$

Constraints



Wigner-Dyson

Characteristic polynomial

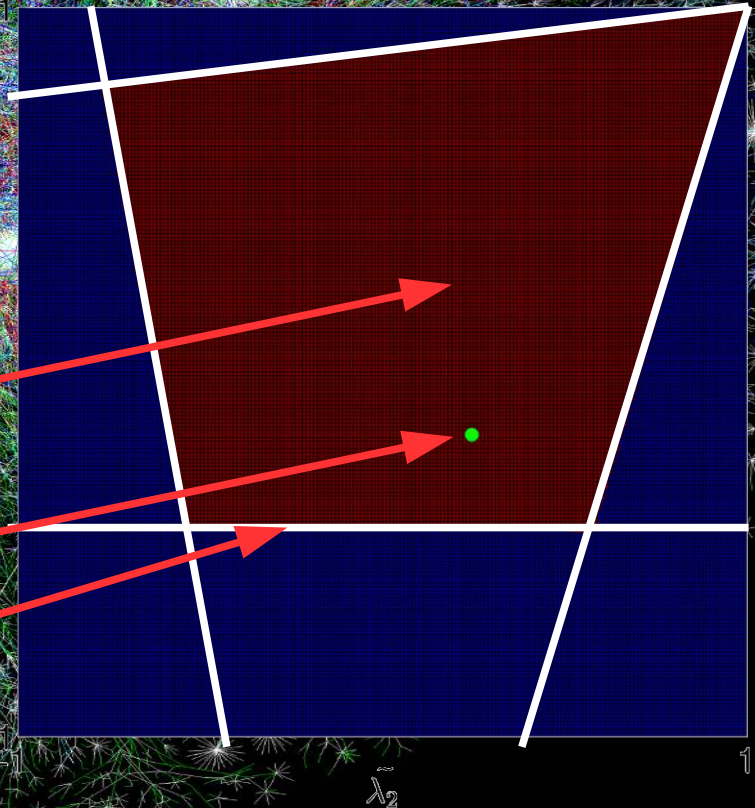
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Admissible $\tilde{T}$

Eigenvalues of T

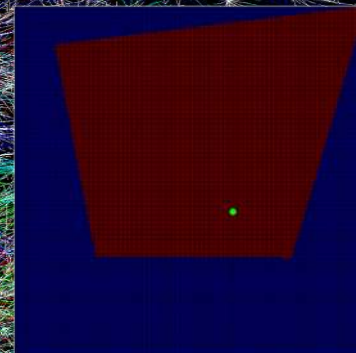
Constraints



Topology of
space of admissible paths

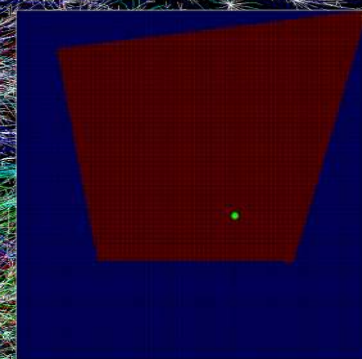
Σ

$\mathcal{X}$

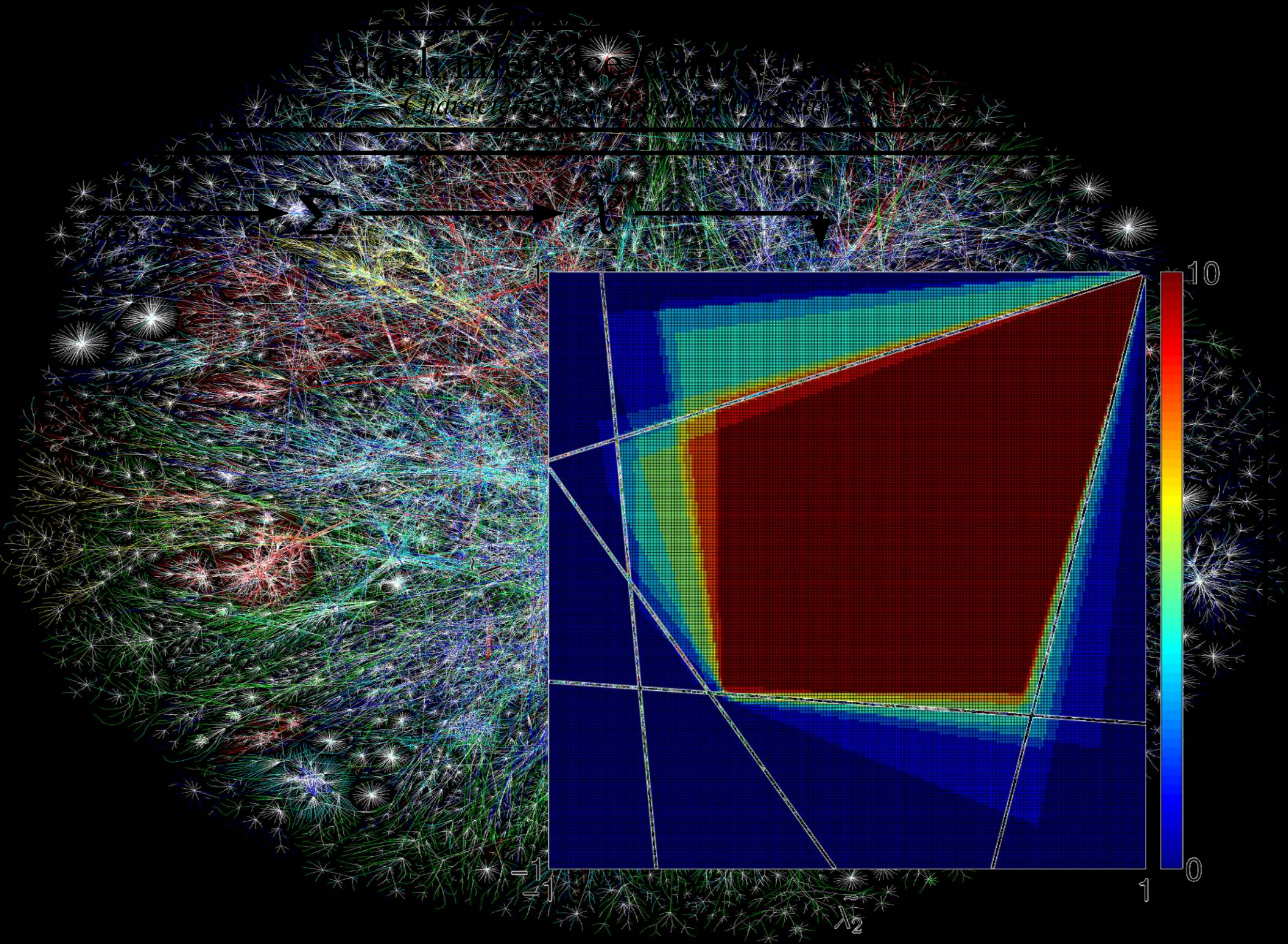


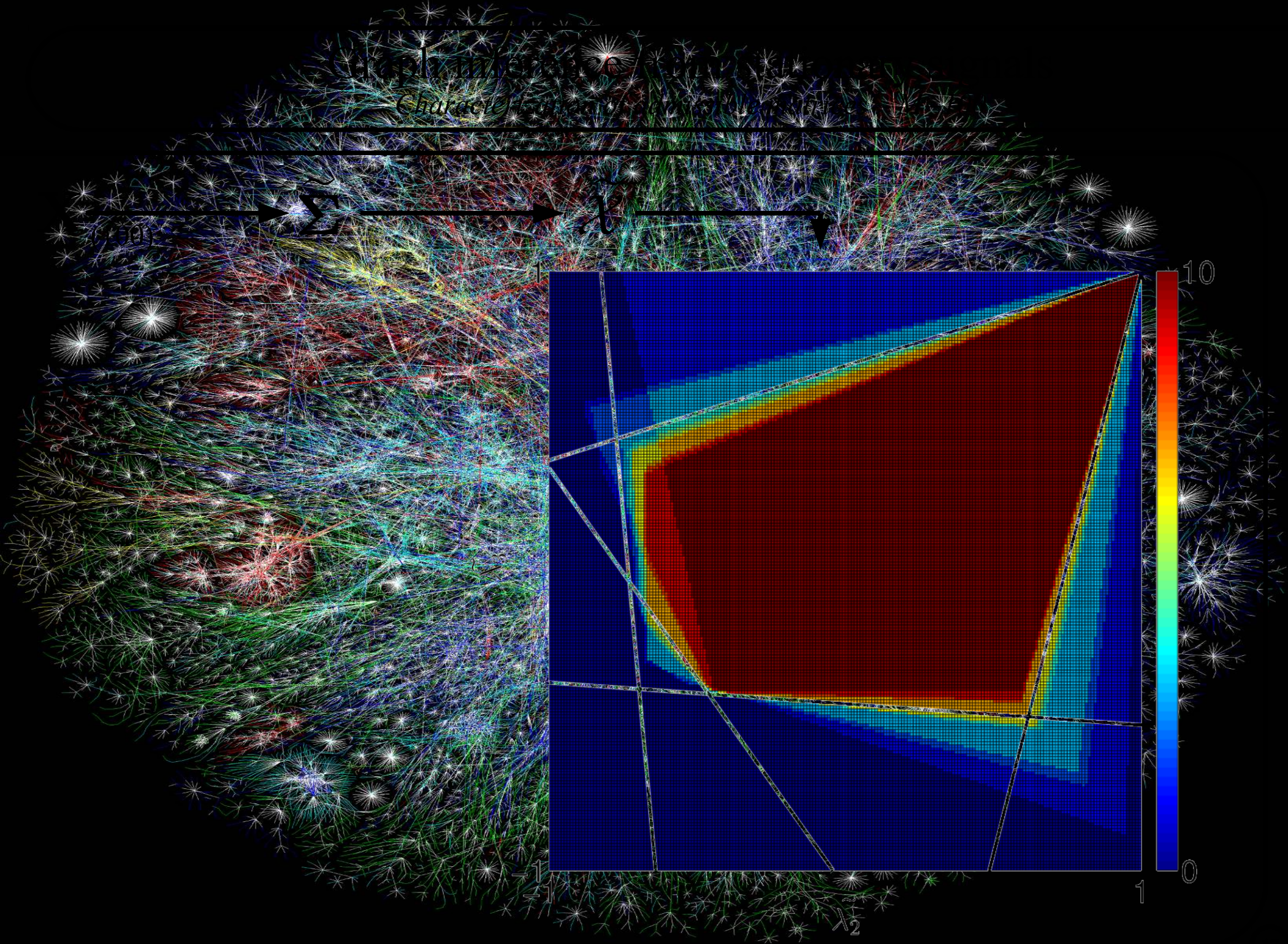
$\mathcal{A}(\mathcal{P})$ is the
 algebra of admissible paths

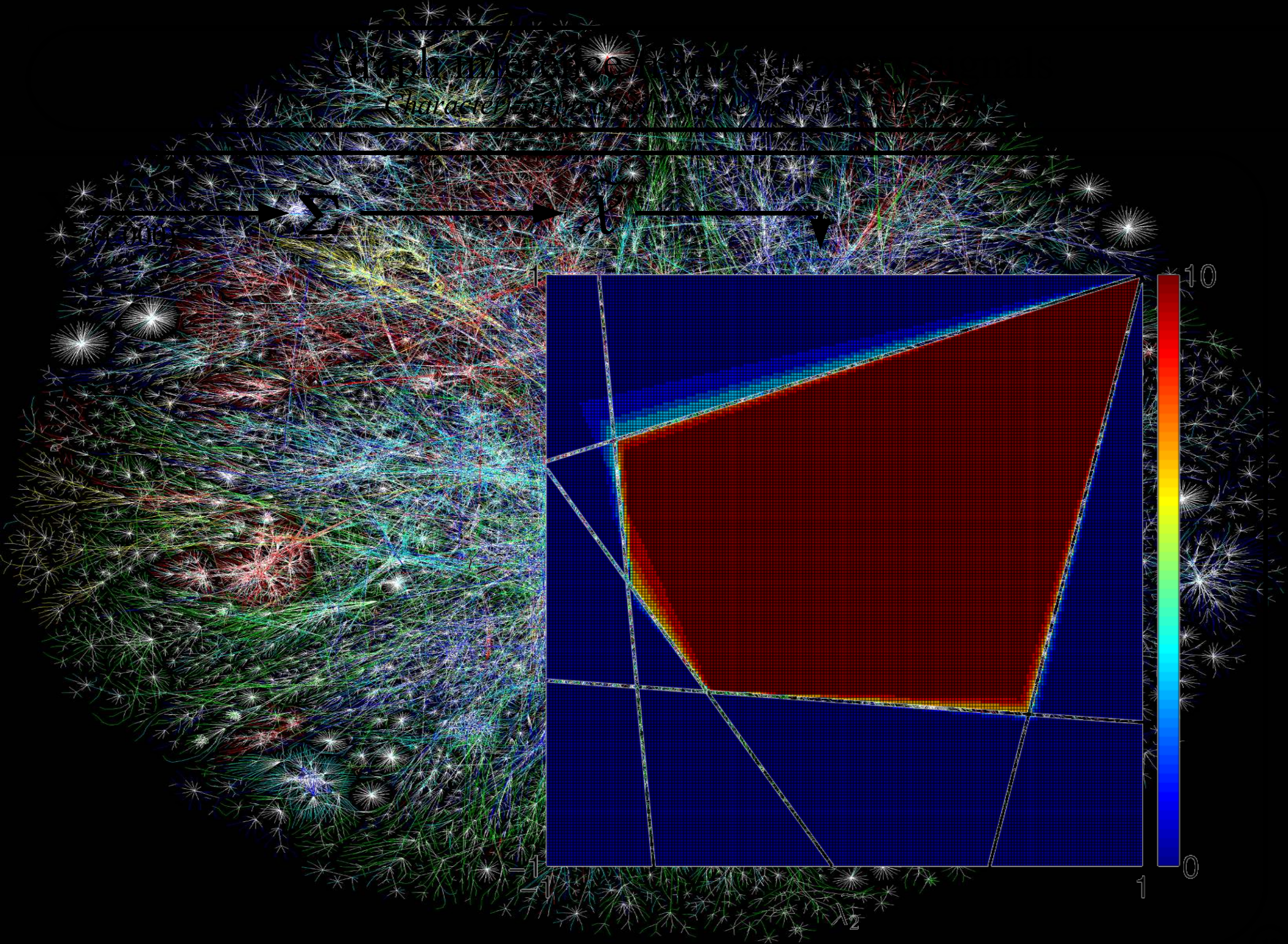
$$\Sigma \longrightarrow \mathcal{X} \longrightarrow \mathcal{Y}$$



$$\Sigma \longrightarrow \tilde{\mathcal{X}} \longrightarrow \mathcal{Y}$$







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Graph inference

Selection of a graph structure from a set of signals

Inference requires selection criteria

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Knowledge of the underlying topology
Knowledge of the diffusion matrix
Knowledge of the signals on the graph
Knowledge of $\tilde{\mathbf{T}}$ (close to) binary matrix

Graph inference

Selection of a graph structure from a set of signals

Inference requires selection criteria

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Our proposed methods

Kalofolias?

Knowledge of the underlying topology
Knowledge of the diffusion matrix
Knowledge of the signals on the graph
Knowledge of a (close to) binary matrix

Graph minimization

Problem of a graph minimization

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph partitioning

Partition of a graph into clusters

Trace of the matrix

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N}$$

$$\sum_{i=1}^N \lambda_i$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Wright

Section of a Transverse Wave

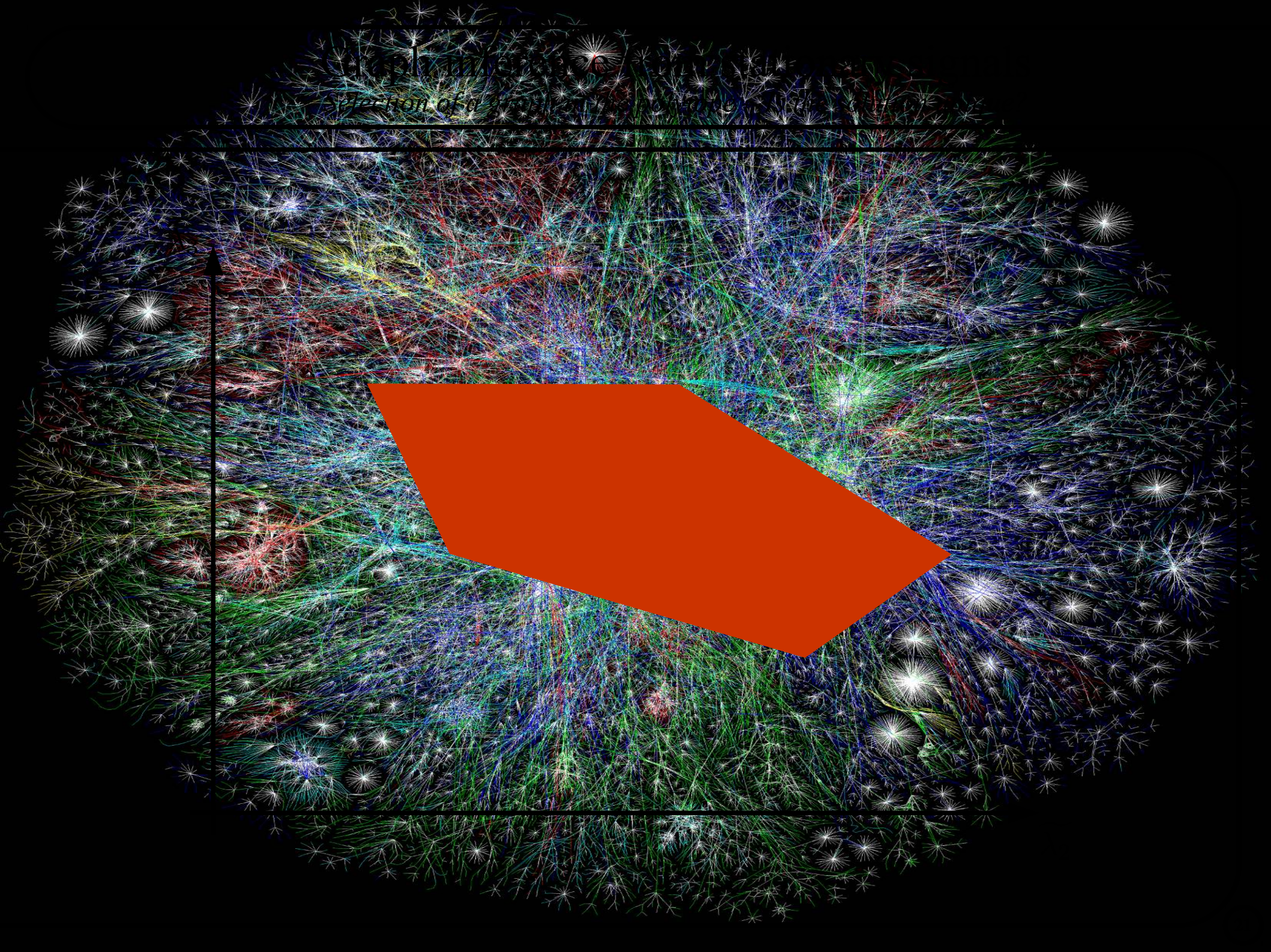
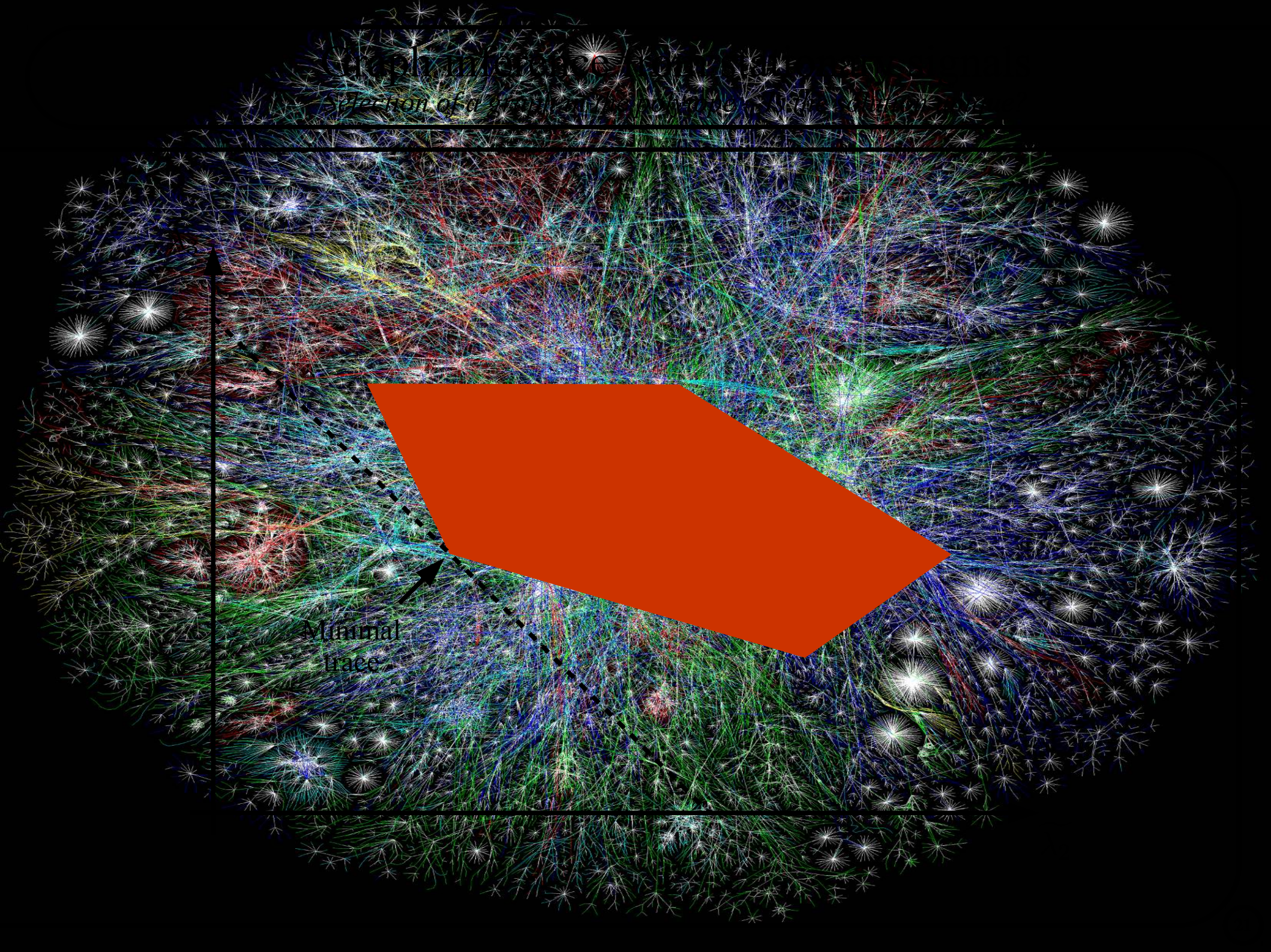


Figure 11

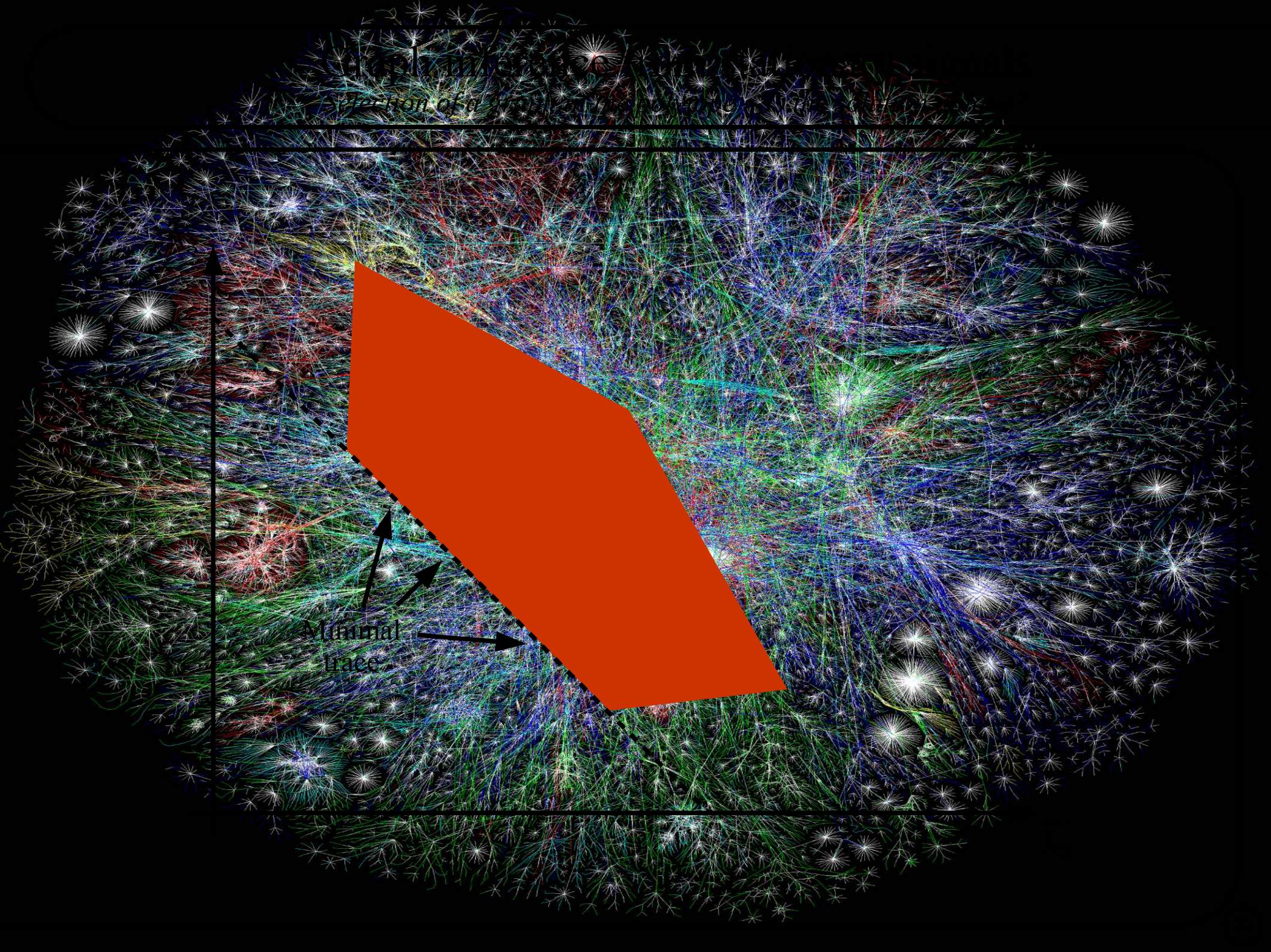
Section of a $\mathbb{P}^2$ with a line



Minimal
trace

Figure 11

Section of a field of force lines



Minimal trace

$$[\lambda_1, \dots, \lambda_N] = \arg \min_{\lambda_1, \dots, \lambda_N} \mathbf{1}_N^T \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^T \mathbf{1}_N$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph partitioning

Partition of a graph into k clusters

ℓ_1 norm

$$[\lambda_1, \dots, \lambda_N] = \arg \min_{\lambda_1, \dots, \lambda_N}$$

$$\mathbf{1}_N^\top \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^\top \mathbf{1}_N$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph mining

Extraction of a graph structure from a network

$\mathbf{X}$

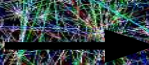
Inference method
(eg. Kalofolias)

$$\hat{\mathbf{T}}_K = \hat{\mathbf{X}}_K \hat{\mathbf{\Lambda}}_K \hat{\mathbf{X}}_K^T$$

Graph with n nodes and m edges
A graph of a galaxy cluster with $n=32$ galaxies

$\mathbf{X}$

Inference method
(eg. Kalofolias)



$$\hat{\mathbf{T}}_K = \hat{\mathbf{X}}_K \hat{\mathbf{\Lambda}}_K \hat{\mathbf{X}}_K^T$$



Not those of the
covariance matrix

Graph mining

Extraction of a graph structure from a network

Inference method
(eg. Kalofolias)

$$\hat{\mathbf{T}}_K = \hat{\mathcal{X}}_K \hat{\mathbf{\Lambda}}_K \hat{\mathcal{X}}_K^T$$

Σ

$$\tilde{\mathcal{X}}_{\tilde{\mathbf{T}}}$$

$$\tilde{\mathbf{T}}_K = \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}} \tilde{\mathbf{A}} \tilde{\mathcal{X}}_{\tilde{\mathbf{T}}}^T$$

Graph mining

Projection of a graph onto a low-dimensional space

Inference method
(eg. Kalofolias)

$$\mathbf{T}_K = \hat{\mathbf{X}}_K \hat{\mathbf{A}}_K \hat{\mathbf{X}}_K^T$$

$\mathbf{X}$

Σ

$$\tilde{\mathbf{X}}_{\mathbf{T}}$$

$$\tilde{\mathbf{T}}_K = \tilde{\mathbf{X}}_{\mathbf{T}} \tilde{\mathbf{A}} \tilde{\mathbf{X}}_{\mathbf{T}}^T$$

"Eigenvalues" of the projected matrix
(not a diagonal matrix)

$$\tilde{\mathbf{A}} = \mathbf{K}_{\mathbf{T}}^T \mathbf{T}_K$$

Graph mining

Extraction of a graph structure from a network

Inference method
(eg. Kalofolias)

$$\mathbf{T}_K = \mathbf{X}_K \mathbf{A}_K \mathbf{X}_K^T$$

$$\Sigma \longrightarrow \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}} \longrightarrow \tilde{\mathbf{T}}_K = \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}} \tilde{\mathbf{A}} \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}^T$$

$$\min \|\mathbf{A} - \text{dense}(\tilde{\mathbf{A}})\|_F \longleftarrow \tilde{\mathbf{A}} = \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}^T \tilde{\mathbf{T}}_K \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}$$

Graph mining

Extraction of a graph structure from a network

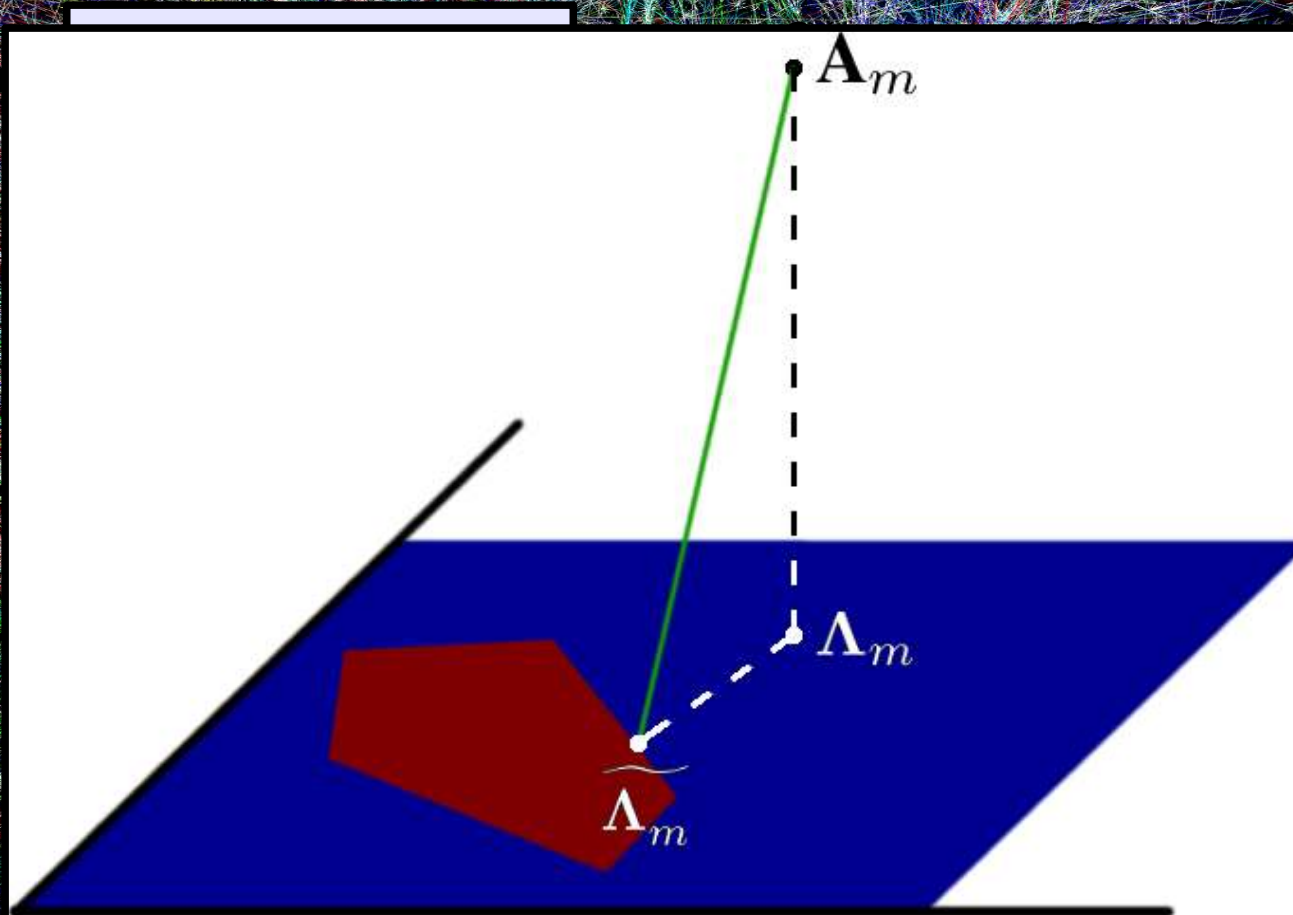
Inference method
(eg. Kalofolias)

$$\mathbf{T}_K = \mathbf{X}_K \mathbf{A}_K \mathbf{X}_K^T$$

$$\Sigma \longrightarrow \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}} \longrightarrow \tilde{\mathbf{T}}_K = \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}} \tilde{\mathbf{A}} \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}^T$$

$$\text{loss} \left\| \mathbf{A} - \text{dense}(\tilde{\mathbf{A}}) \right\|_F \longleftarrow \tilde{\mathbf{A}} = \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}^T \tilde{\mathbf{T}}_K \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}$$

Figure 1: A diagram illustrating the geometry of a system. The diagram shows a cross-section of a system with a central region (red) and surrounding regions (blue and white). The central region is labeled Λ_m . The surrounding regions are labeled Λ_m and Λ_m . The diagram also shows a coordinate system with axes x and y .



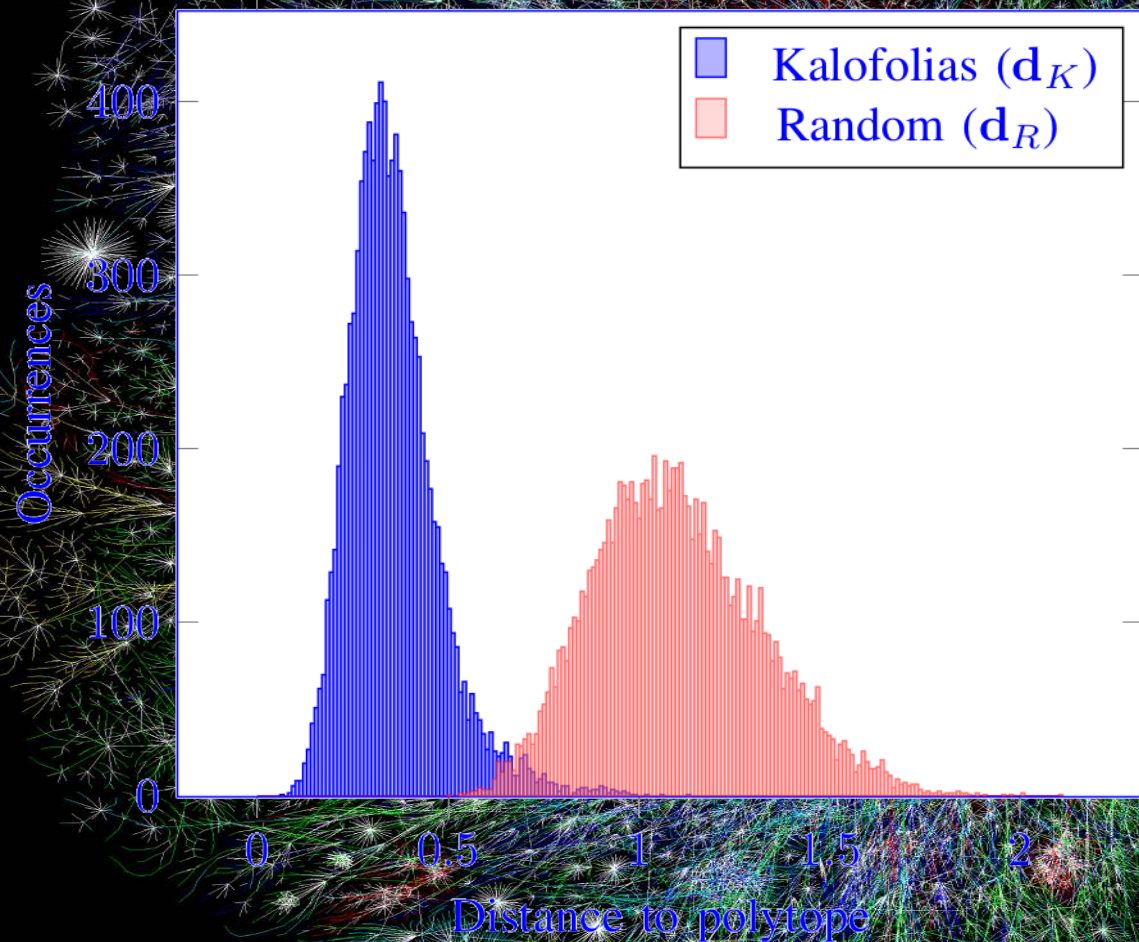


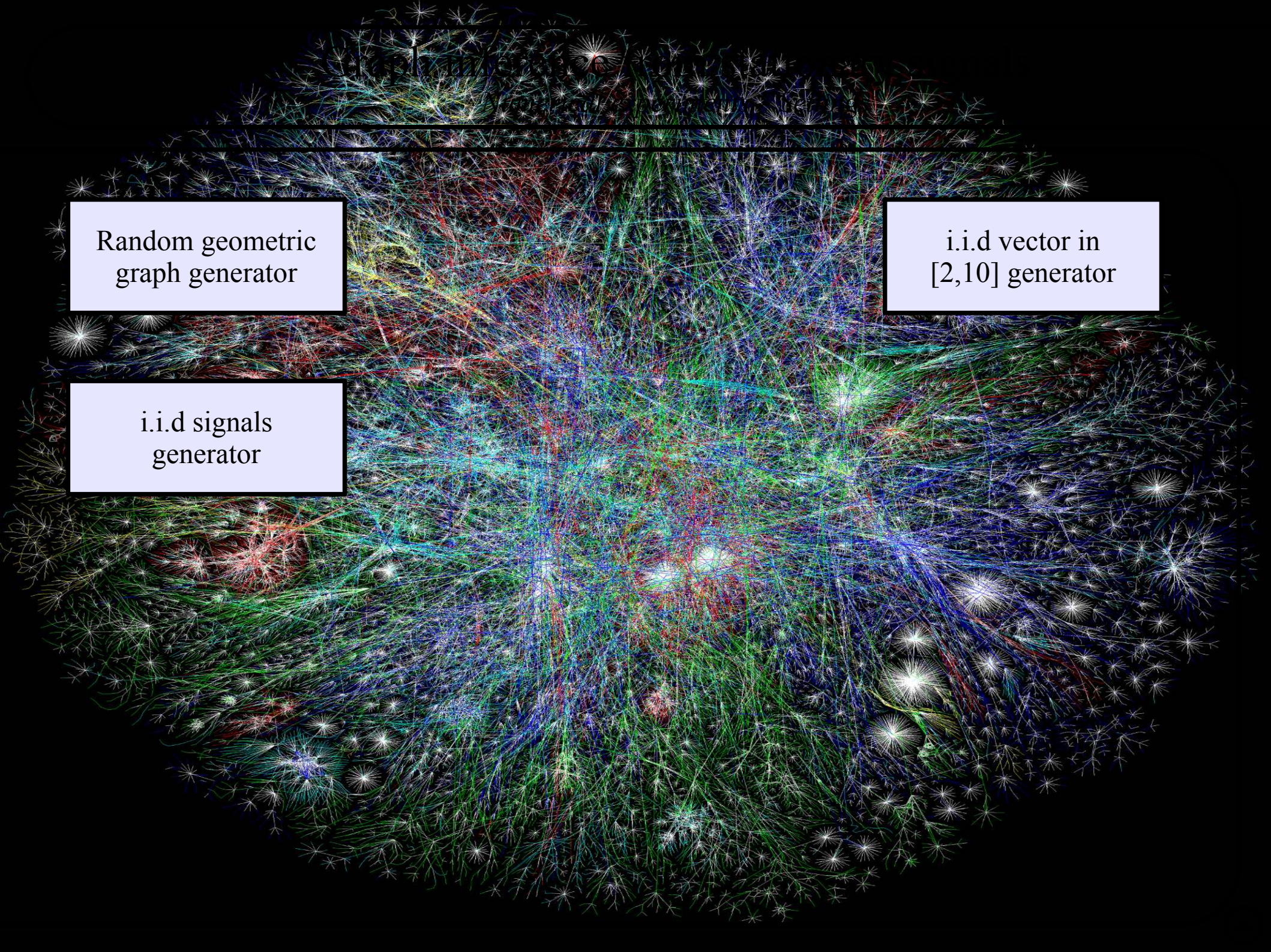
Figure 8: Histogram of the number of points closer to the ground truth polytope than to the closest polytope of the other signals. The distance was computed using either the method from Kalofolias [7] (d_K), or the method from [10] (d_R). Distances are grouped in bins of size 0.01. The red distribution corresponds to the occurrences of random geometric graphs not matched with the ground truth signals.

► "Recent" work on graph inference from stationary signals

- * Introduction / problem formulation
- * Related work
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- * Strategies for selecting a graph in the polytope
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- * Tropical graph signal processing (?)



Random geometric
graph generator

i.i.d vector in
[2,10] generator

i.i.d signals
generator

Random geometric
graph generator

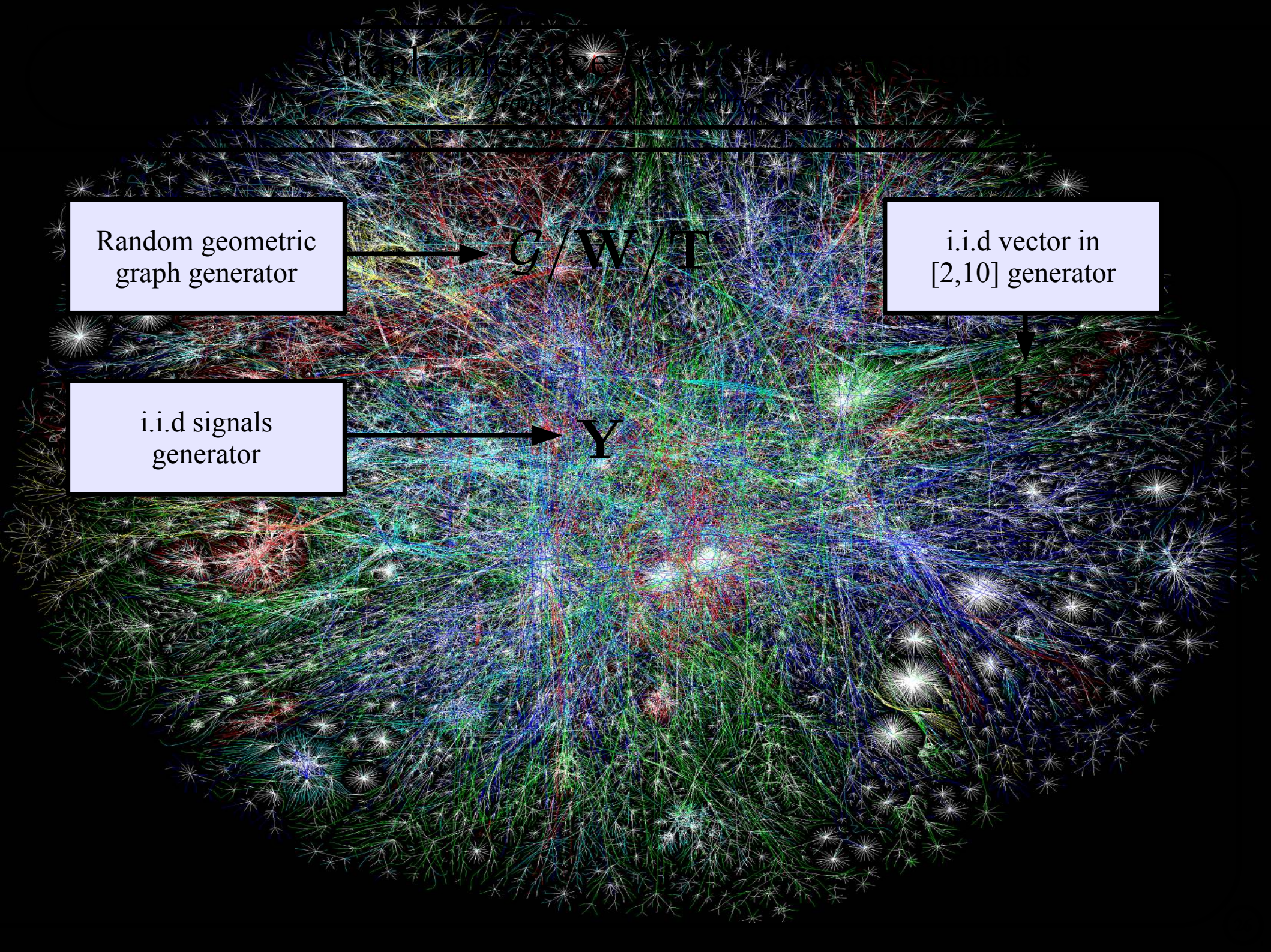
$\mathcal{G}/\mathcal{W}/\mathcal{T}$

i.i.d vector in
[2,10] generator

i.i.d signals
generator

$\mathbf{Y}$

$\mathbf{k}$



Random geometric
graph generator

$\mathcal{G}/\mathbf{W}/\mathbf{T}$

i.i.d vector in
[2,10] generator

i.i.d signals
generator

$\mathbf{Y}$

$\mathbf{k}$

$$\forall i \in \{1, \dots, M\} : \mathbf{X}(i) = \mathbf{T}^{\mathbf{k}(i)} \mathbf{Y}(i)$$

Random geometric
graph generator

$\mathcal{G}/\mathbf{W}/\mathbf{T}$

i.i.d vector in
[2,10] generator

i.i.d signals
generator

$\mathbf{Y}$

$\mathbf{k}$

$$\forall i \in \{1, \dots, M\} : \mathbf{X}(i) = \mathbf{T}^{\mathbf{k}(i)} \mathbf{Y}(i)$$

Inference
algorithm

$\hat{\mathbf{T}}$

Algorithms

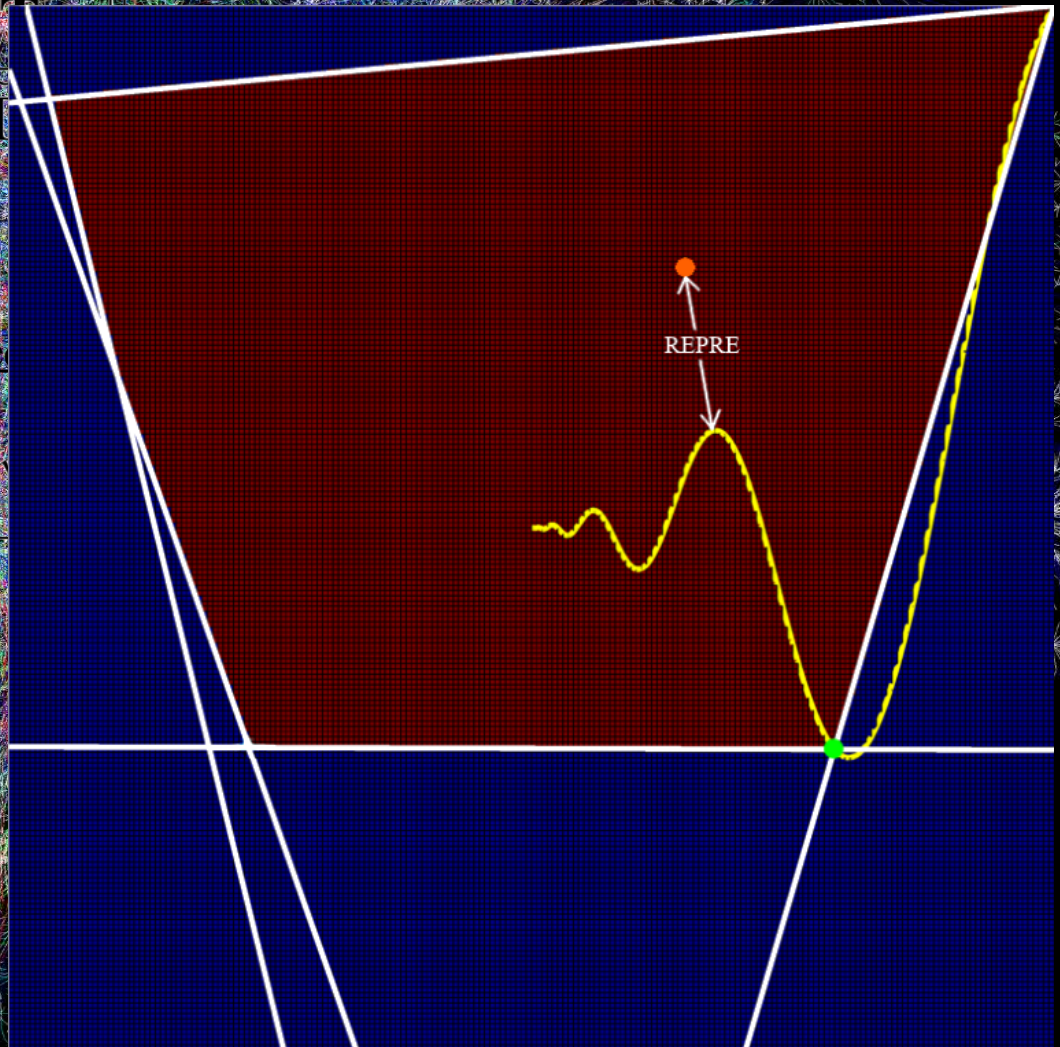
Numerical experiments

Matrix Λ constructed Entry

$$\text{MEPRE}(\mathbf{T}, \mathbf{T}) = \frac{1}{N} \left\| \mathbf{T} \right\|_F^2$$

Construction Error of the Powered Retrieve

$$\text{REPRE}(\Lambda, \Lambda) = \min_{\Lambda^k \in \Lambda} \frac{1}{N} \left\| \Lambda^k \right\|_F^2$$



Graph metrics

Empirical precision and recall

Model metrics evaluated on thresholded binarized matrices

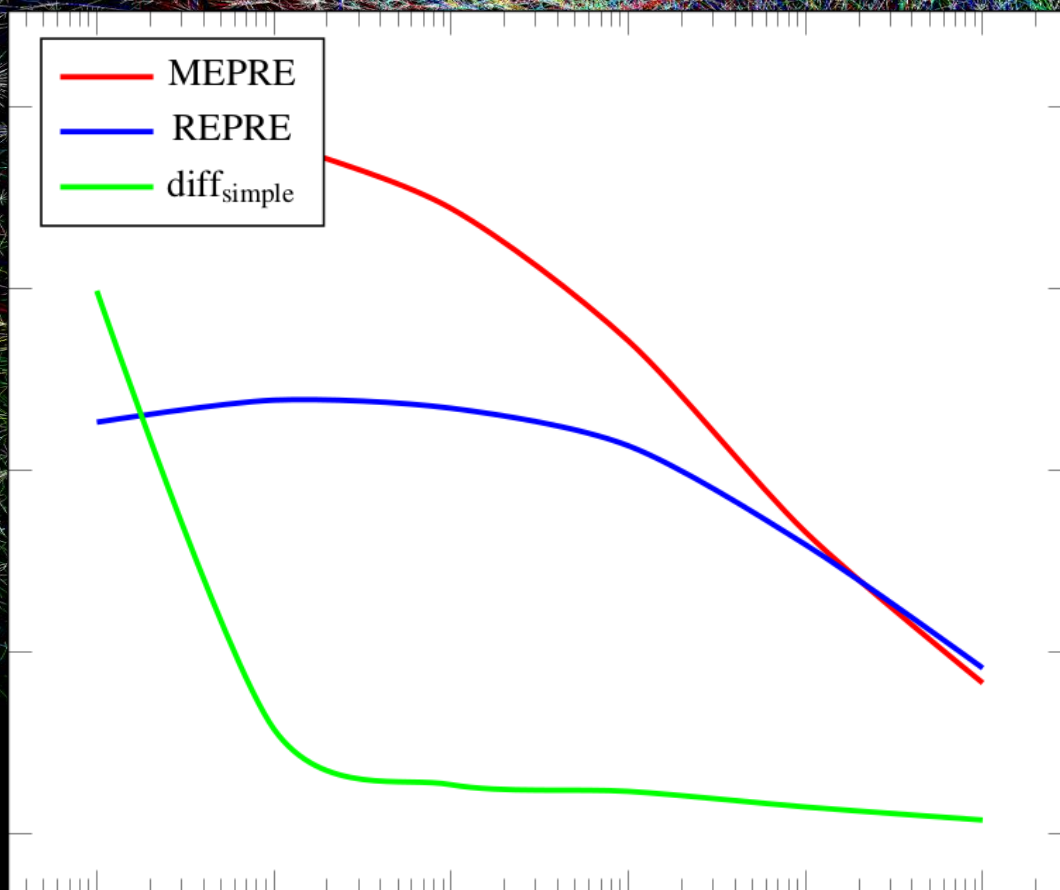
$$\text{precision}(\mathbf{T}, \mathbf{T}_{thr}) \triangleq \frac{\#\{(i, j) | \mathbf{T}_{thr}(i, j) > 0 \text{ and } \mathbf{T}(i, j) > 0\}}{\|\mathbf{T}_{thr}\|_{0,1}}.$$

$$\text{recall}(\mathbf{T}, \mathbf{T}_{thr}) \triangleq \frac{\#\{(i, j) | \mathbf{T}_{thr}(i, j) > 0 \text{ and } \mathbf{T}(i, j) > 0\}}{\|\mathbf{T}\|_{0,1}}.$$

$$\text{F-measure}(\mathbf{T}, \mathbf{T}_{thr}) \triangleq 2 \frac{\text{precision}(\mathbf{T}, \mathbf{T}_{thr}) \text{recall}(\mathbf{T}, \mathbf{T}_{thr})}{\text{precision}(\mathbf{T}, \mathbf{T}_{thr}) + \text{recall}(\mathbf{T}, \mathbf{T}_{thr})}.$$

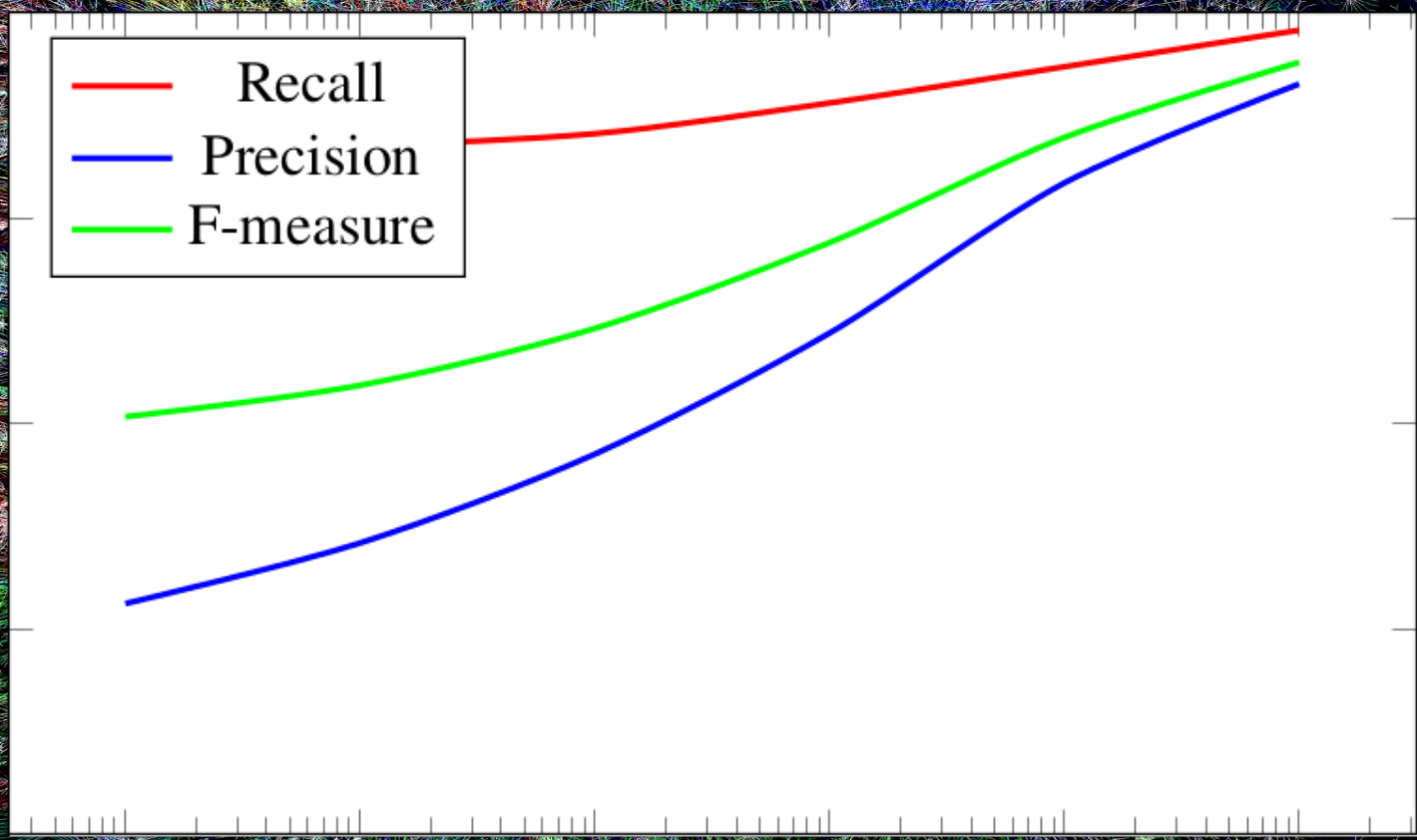
Visualizing Symplectic embeddings

$$\text{diff}_{\text{simple}}(\mathbf{T}, \mathbf{T}) \triangleq \frac{1}{N} \left(\text{Tr}(\mathbf{T}) - \text{Tr}(\mathbf{I}) \right)$$

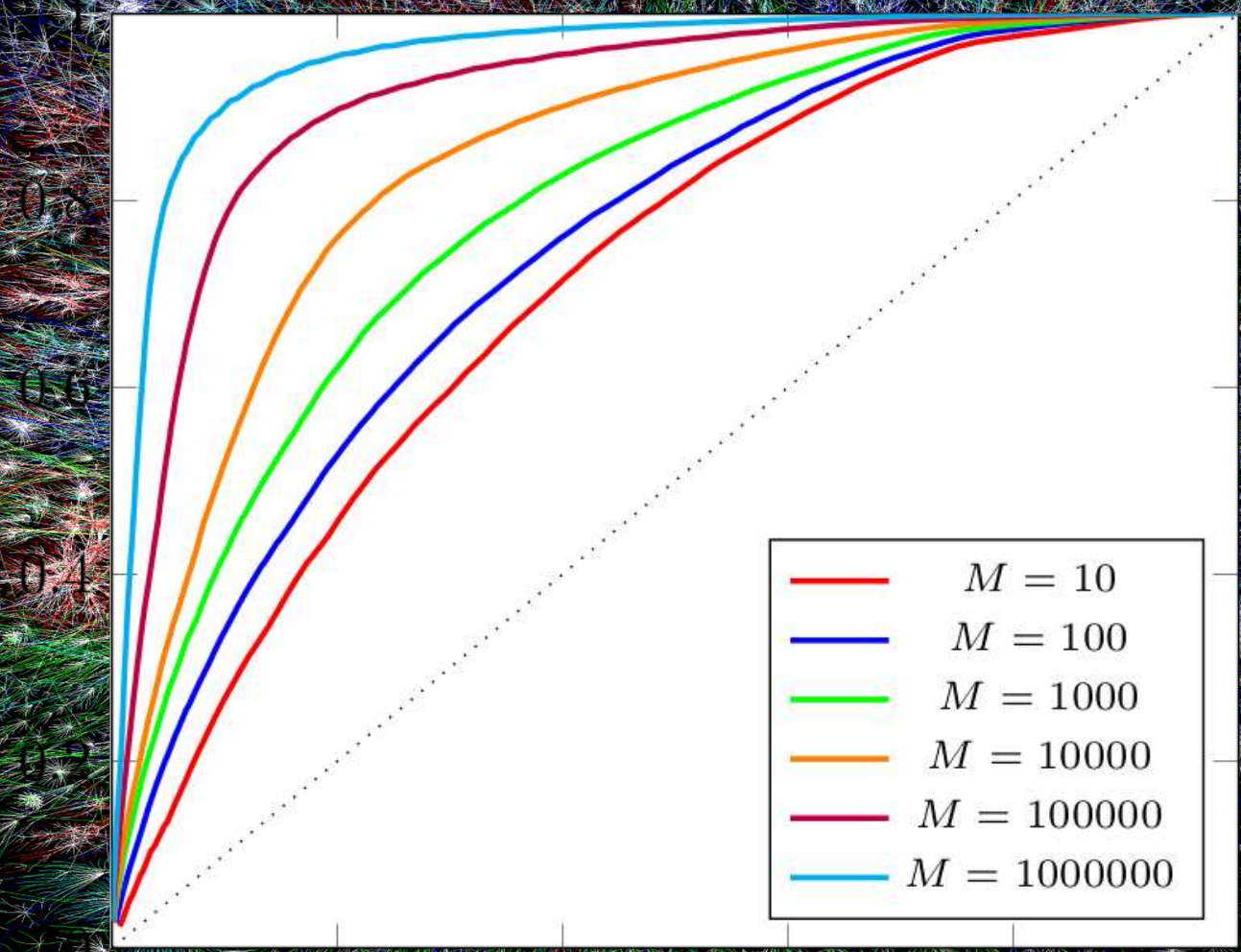


ViPhy 170

Empirical experiments



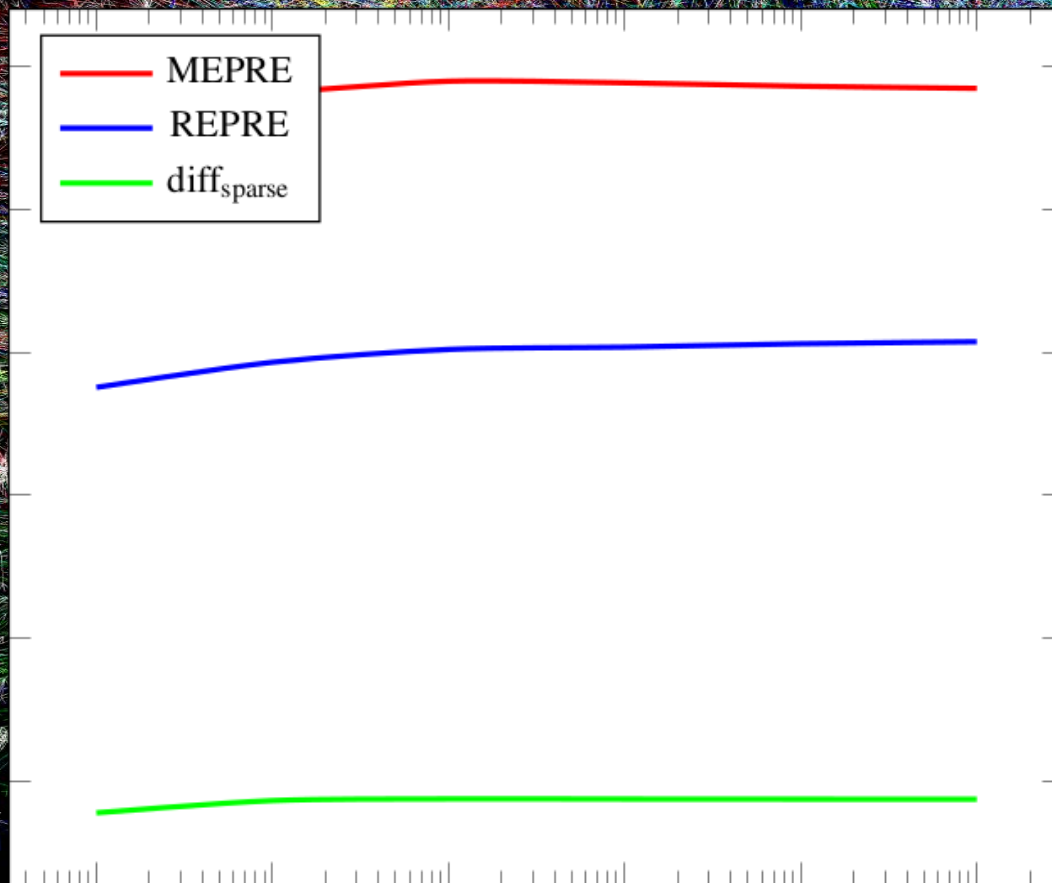
Empirical



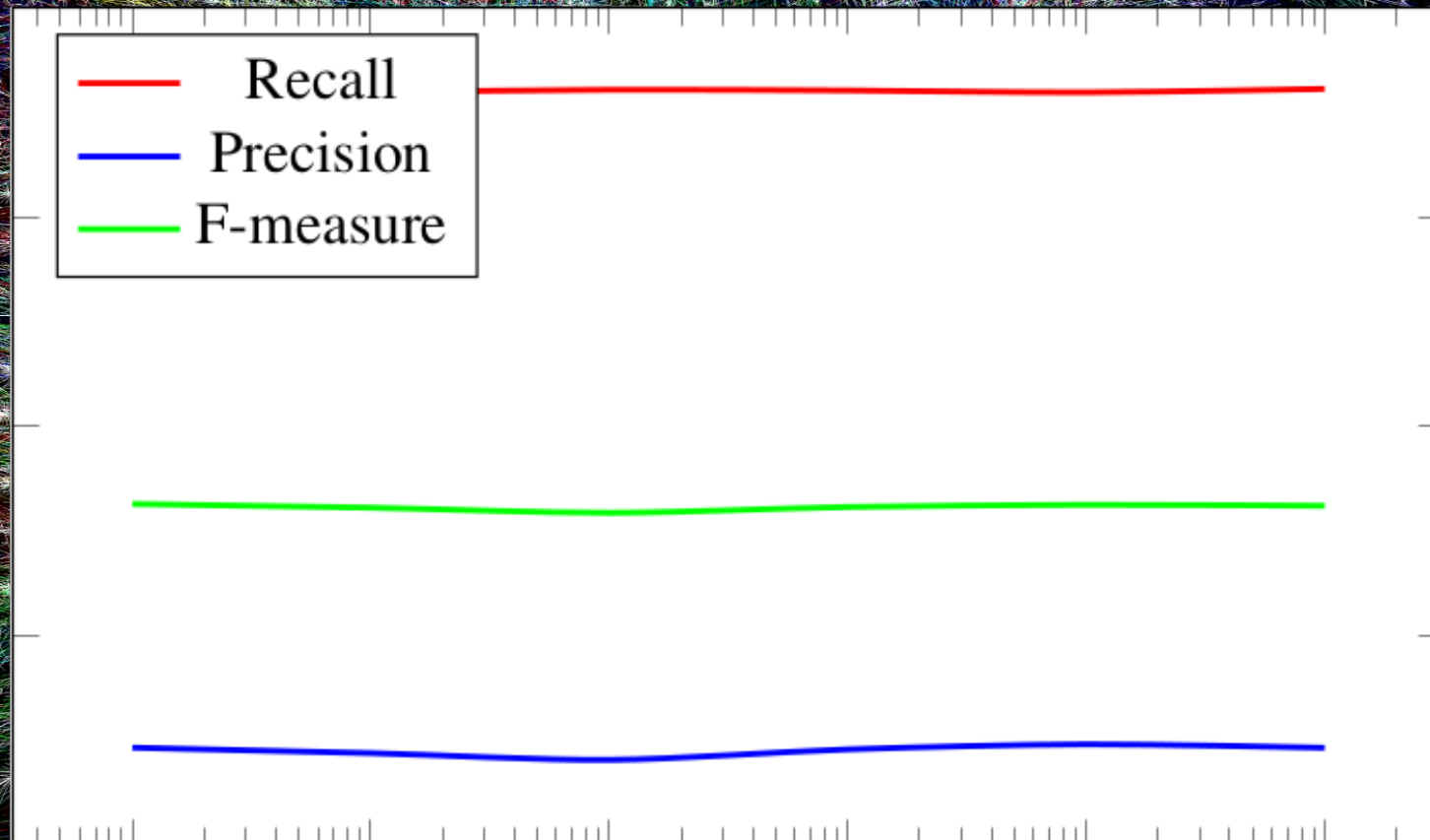
Graph metrics

Symmetrical experiments

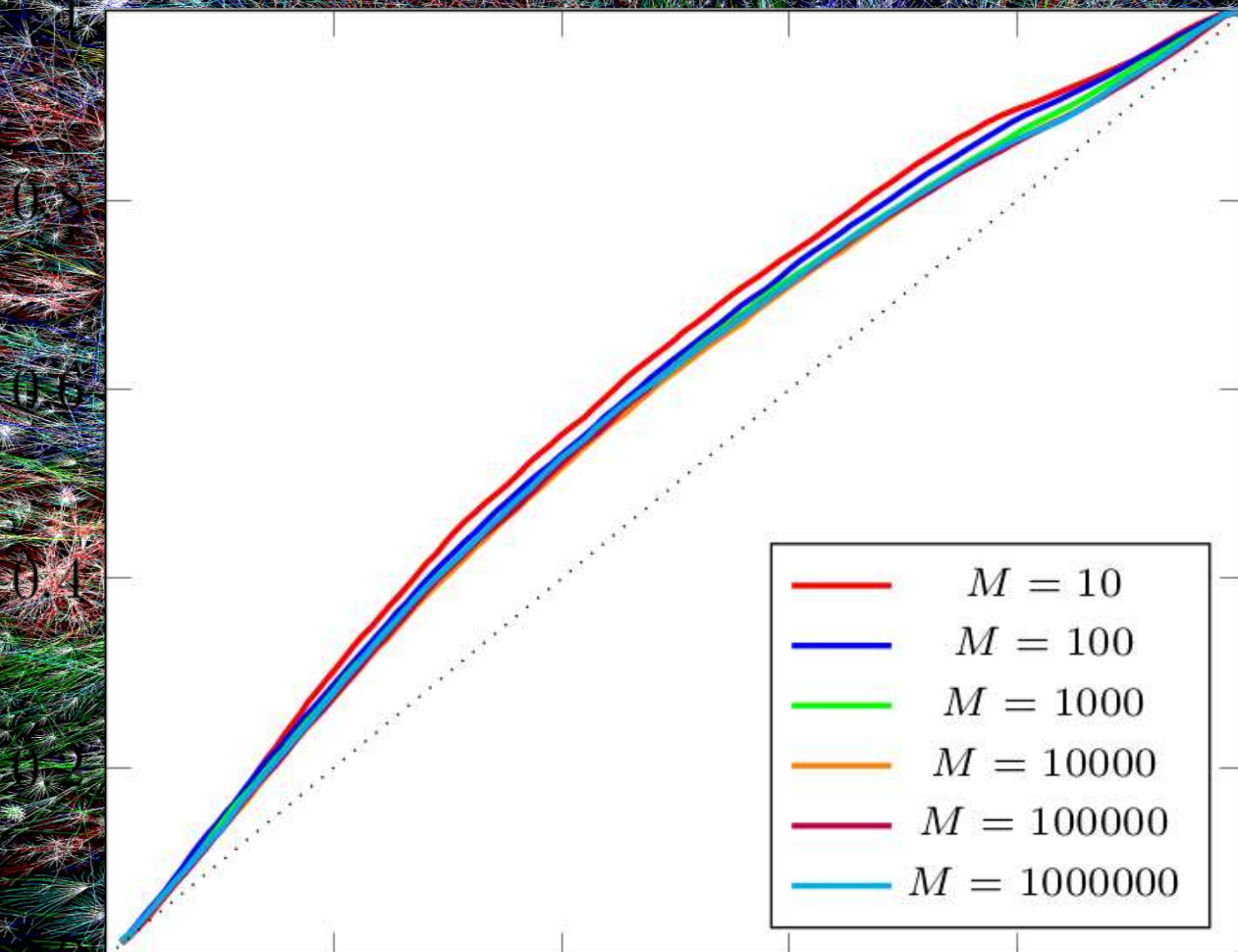
$$\text{diff}_{\text{sparse}}(\mathbf{T}, \mathbf{\hat{T}}) \triangleq \frac{1}{N^2} \left(\|\mathbf{T}\|_{1,1} - \|\mathbf{\hat{T}}\|_{1,1} \right)$$



Viola-Jones Empirical

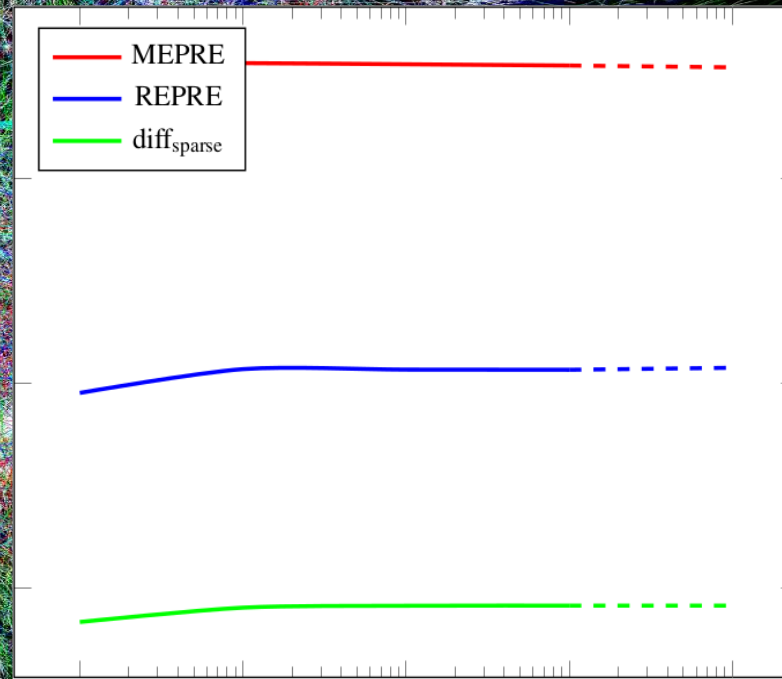
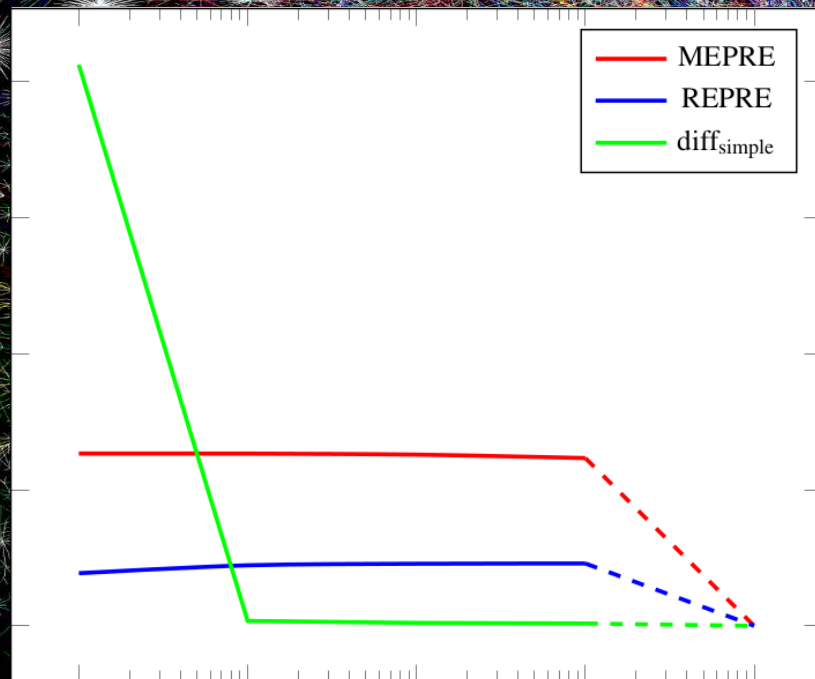


Empirical Symmetrical exponential



Diffusion

Numerical Methods



► "Recent" work on graph inference from stationary signals

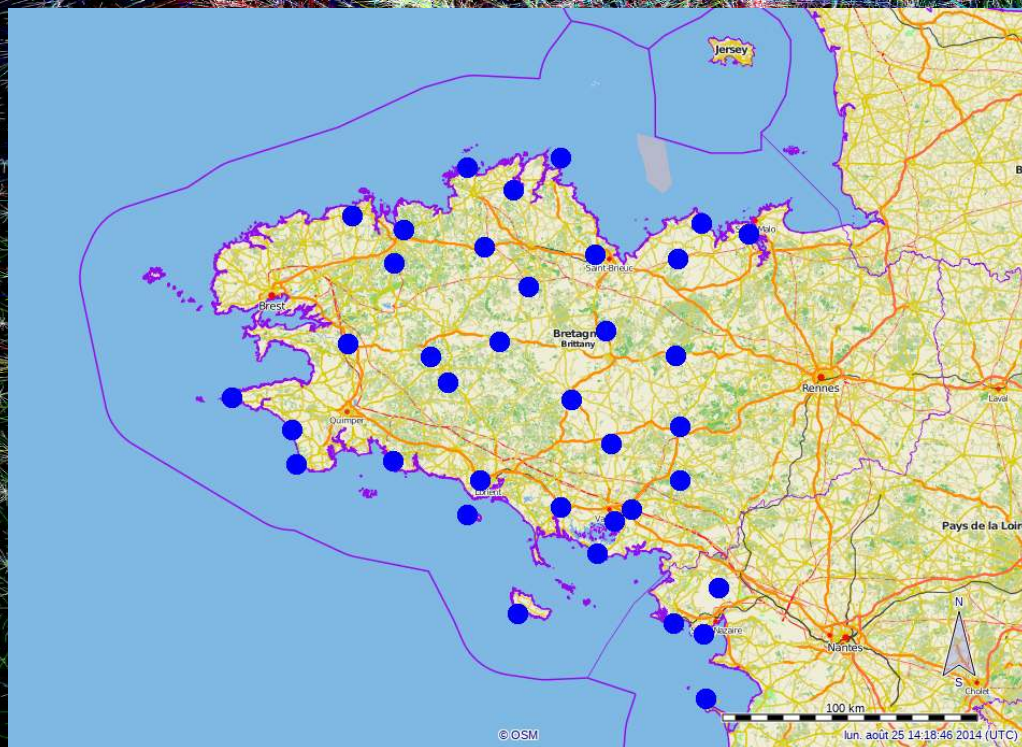
- * Introduction / problem formulation
- * Related work
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► Ongoing and future work

- * Application of graph inference to signals classification
- * Tropical graph signal processing (?)

Graph mining

Real life problems



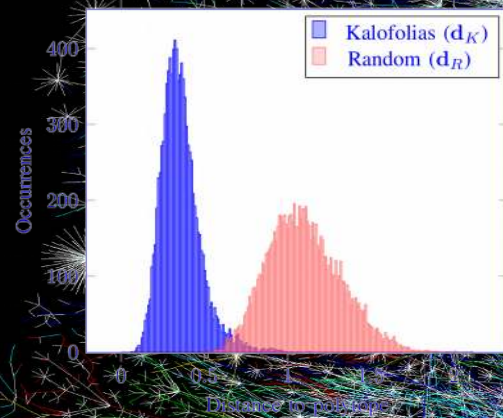
37 stations after preprocessing

744 observations per station

Our Barbary vs X Grignon 2017

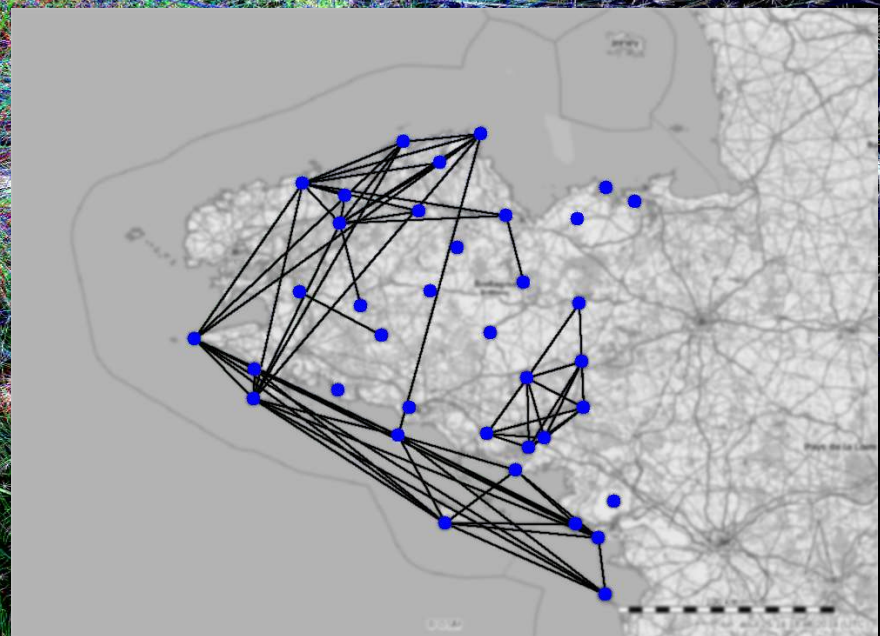
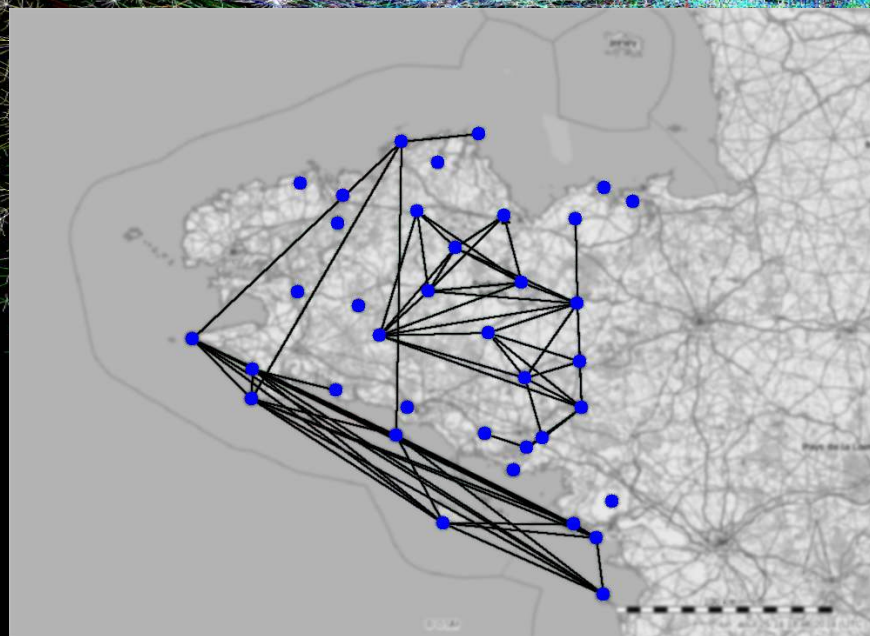
Wasp movement

real-life experiment vs. *computer simulation*



Kalofolias

"Regularized" Kalofolias



Graphical lasso

Real-life example

	polytope	$L_{1,1}$	Tr	$S(\mathbf{X})$
Simple	✓	36.9974	0.0013	0.0551
Sparse	✓	36.9971	0.9093	0.0585
Kalofolias [7]	0.0313	36.9979	0	0.0751
Kalofolias closest	✓	36.9974	0.0298	0.0548
Segarra <i>et al.</i> [39]	0.0062	36.9993	$1.97 \cdot 10^{-5}$	0.0245
Segarra <i>et al.</i> closest	✓	36.9974	0.0046	0.0551
Graphical lasso [25]	1.3730	35.3539	13.4977	32.8421
Graphical lasso closest	✓	36.9984	13.3584	0.0335

► "Recent" work on graph inference from stationary signals

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► Ongoing and future work

► Application of graph inference to signals classification
► Graphical graph signal processing (2)

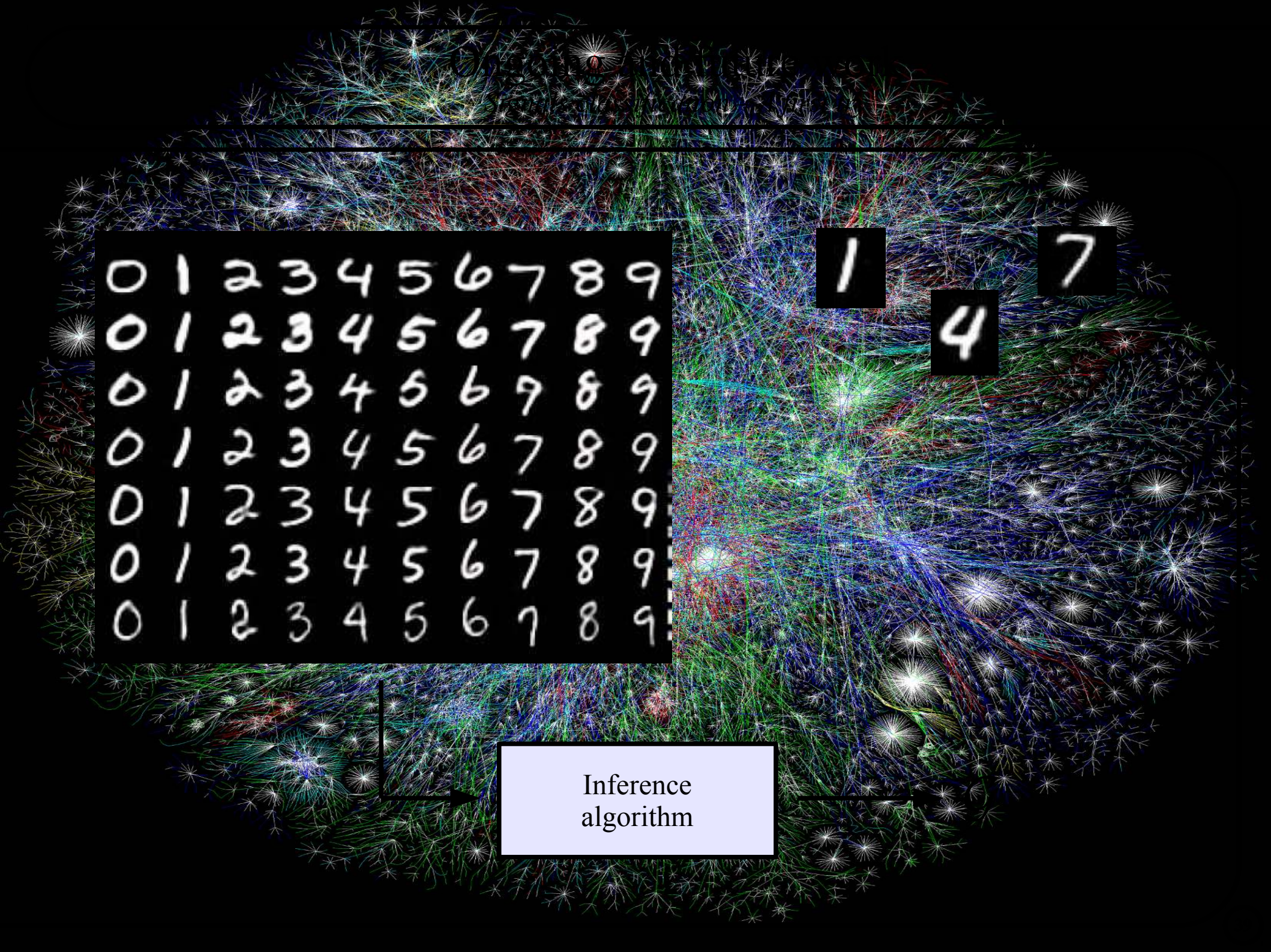
► "Recent" work on graph inference from stationary signals

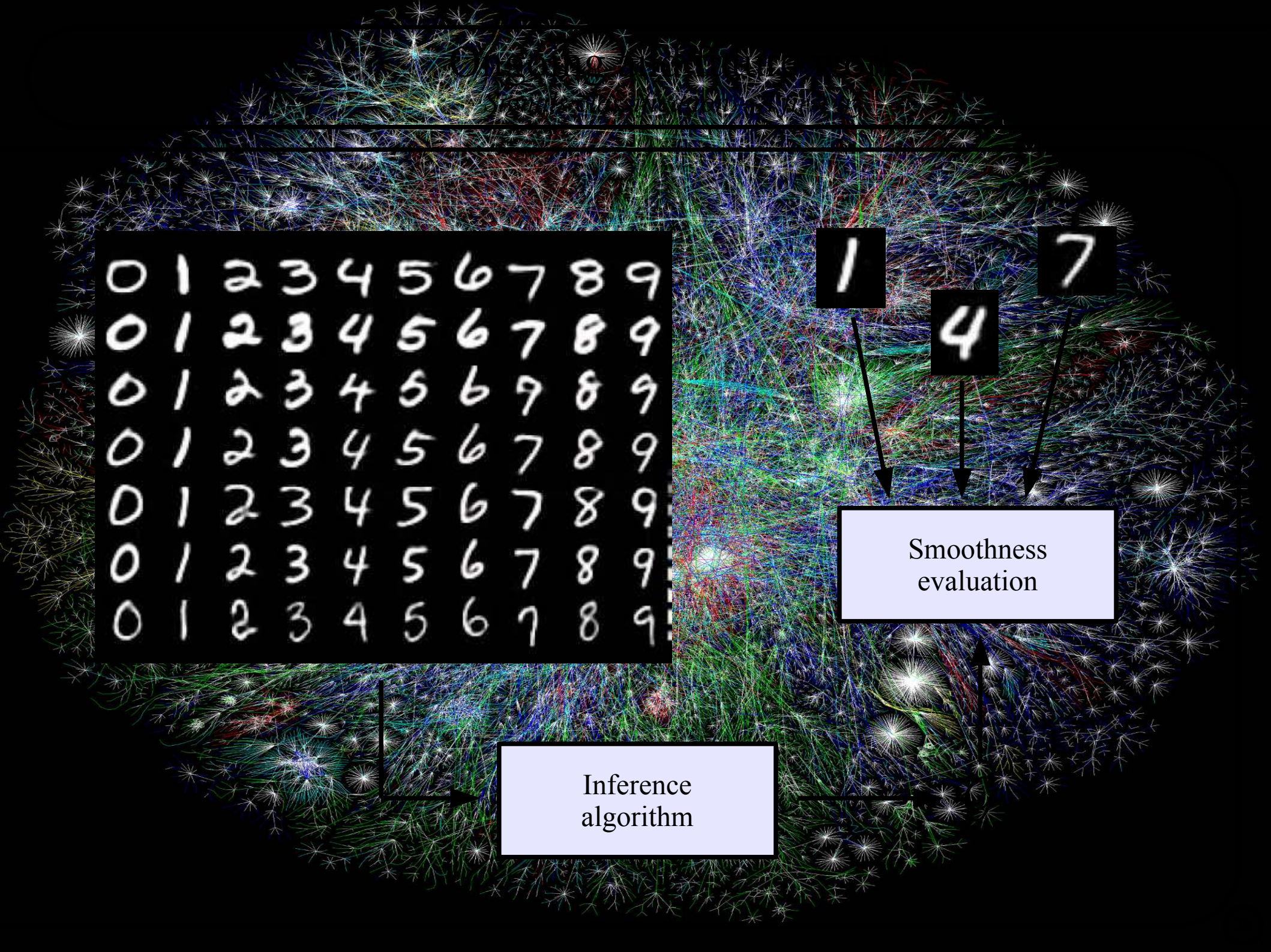
- * Introduction / problem formulation
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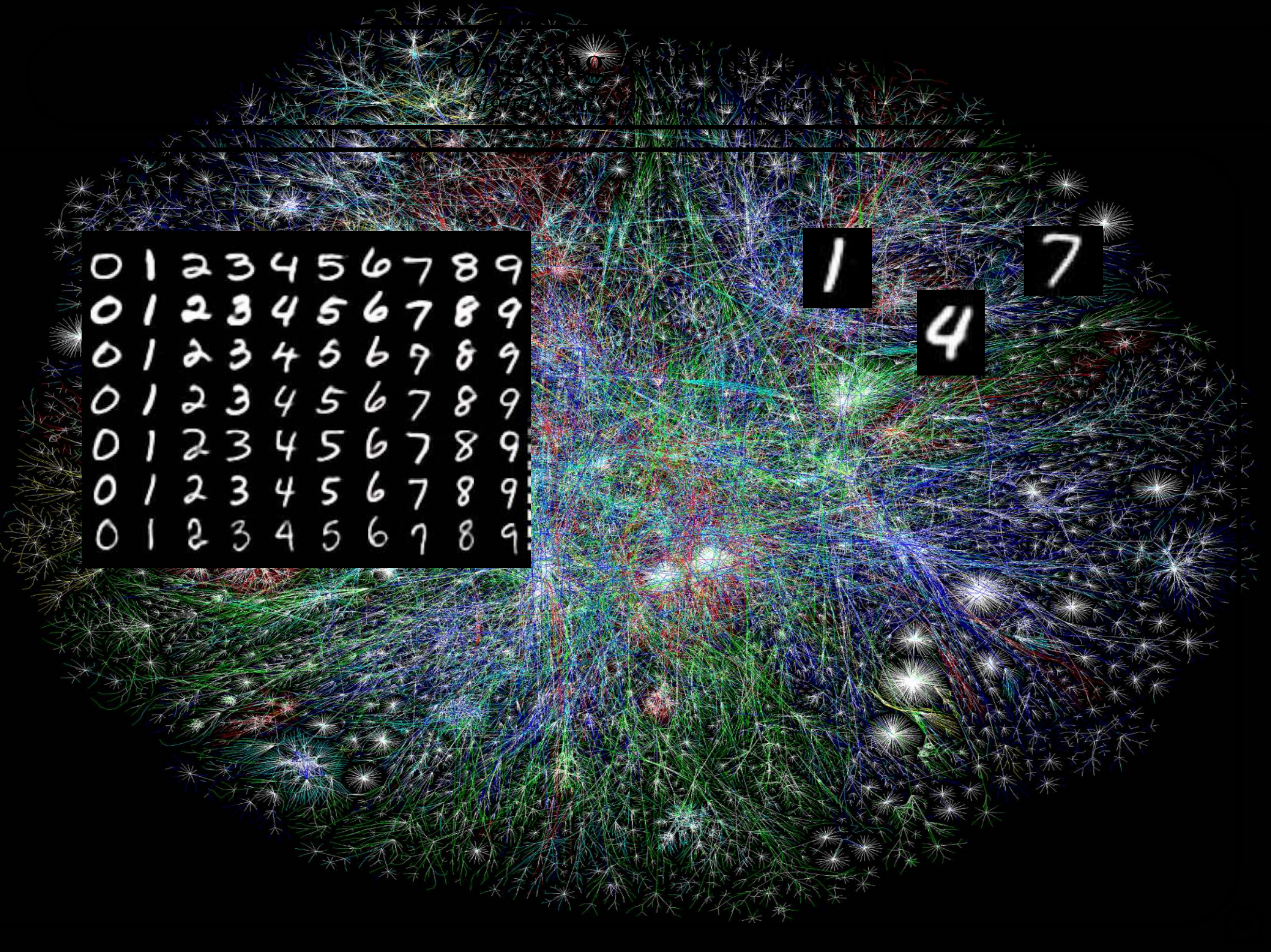
► Ongoing and future work

- * Application of graph inference to signals classification
- * Tropical graph signal processing (?)







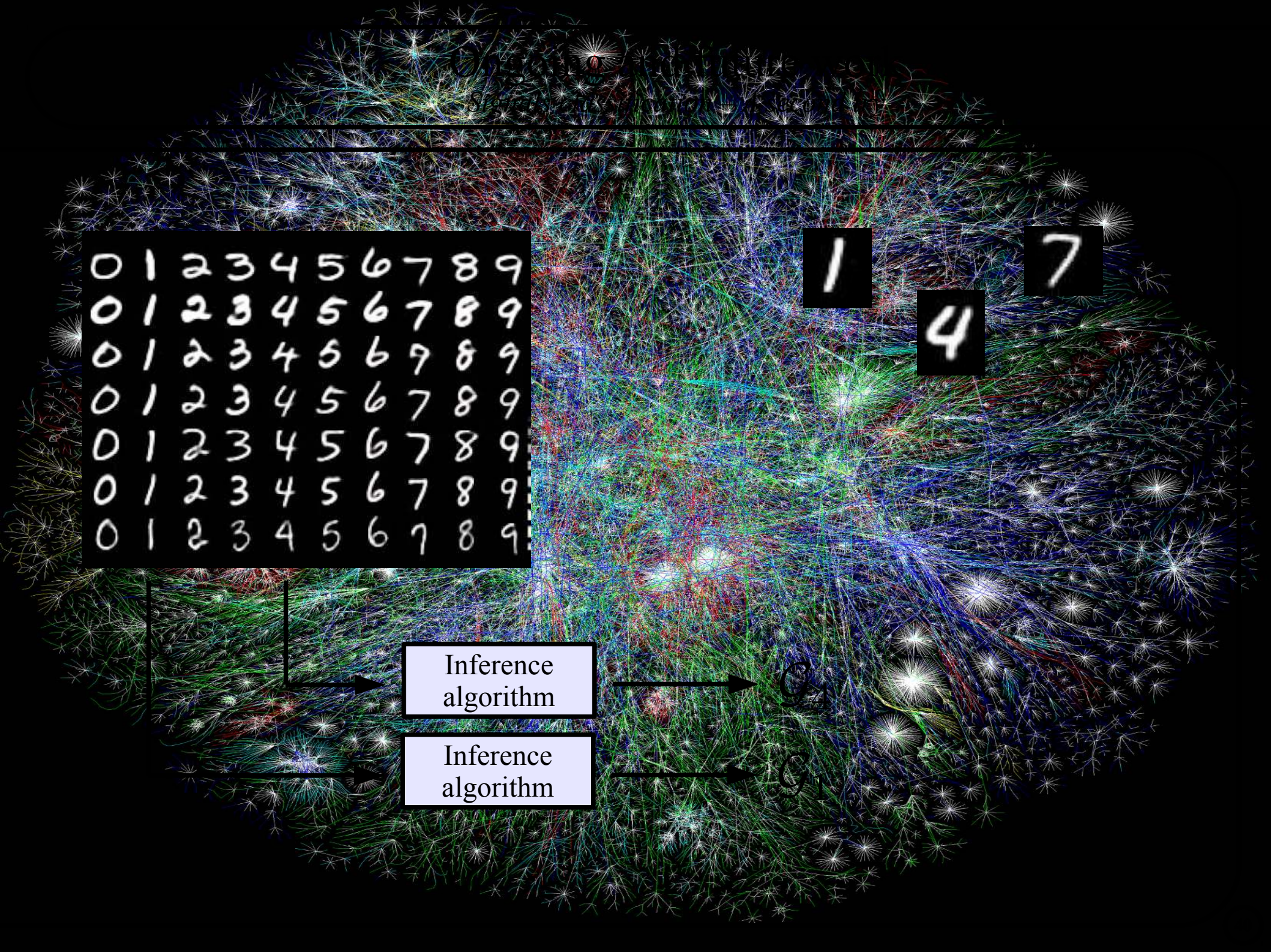


0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	9	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

1

7

4



0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	9	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

1

4

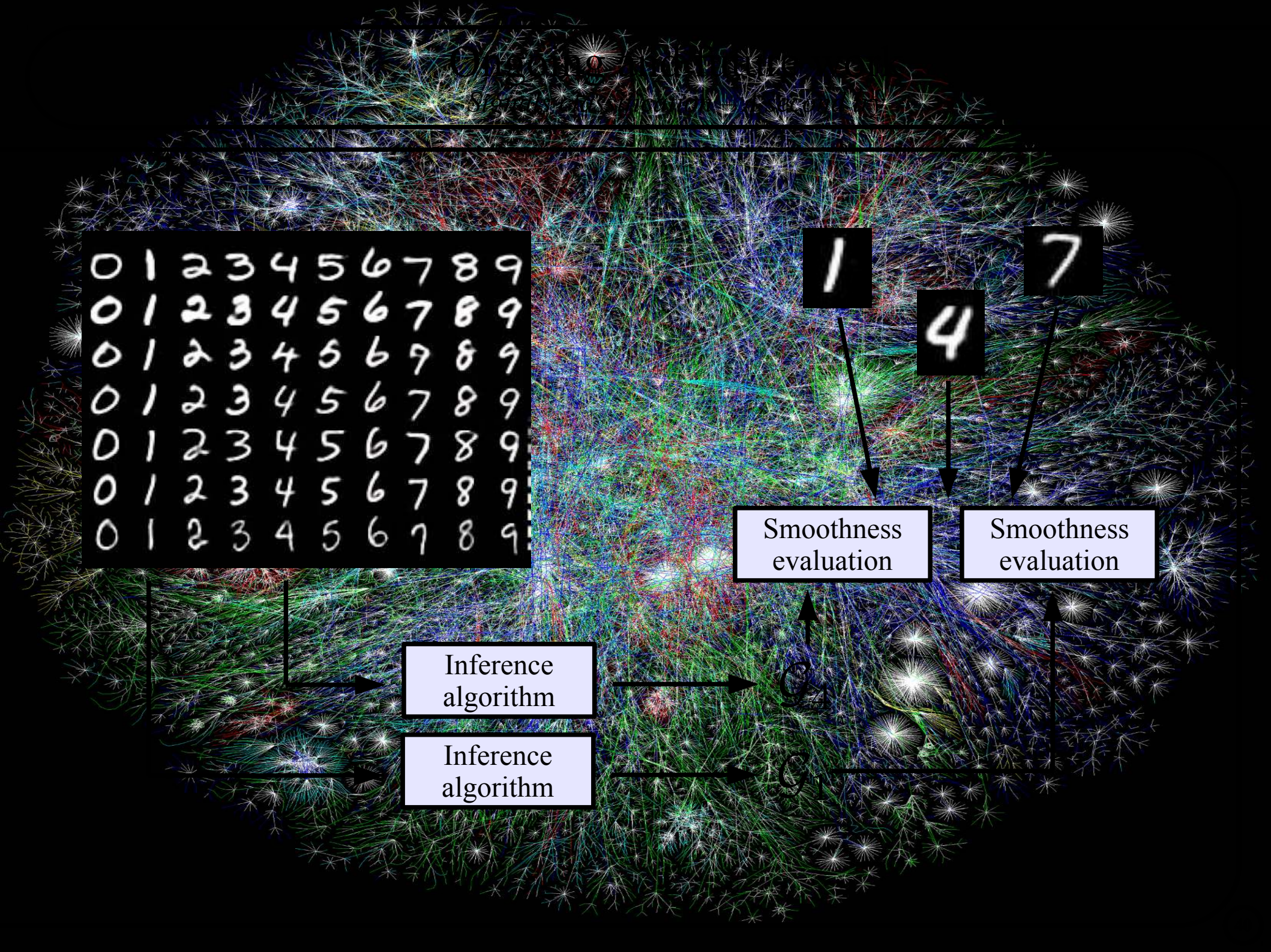
7

Inference
algorithm

Inference
algorithm

$\mathcal{G}_1$

$\mathcal{G}_1$



0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	9	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

Inference
algorithm

Inference
algorithm

Smoothness
evaluation

Smoothness
evaluation

1

4

7

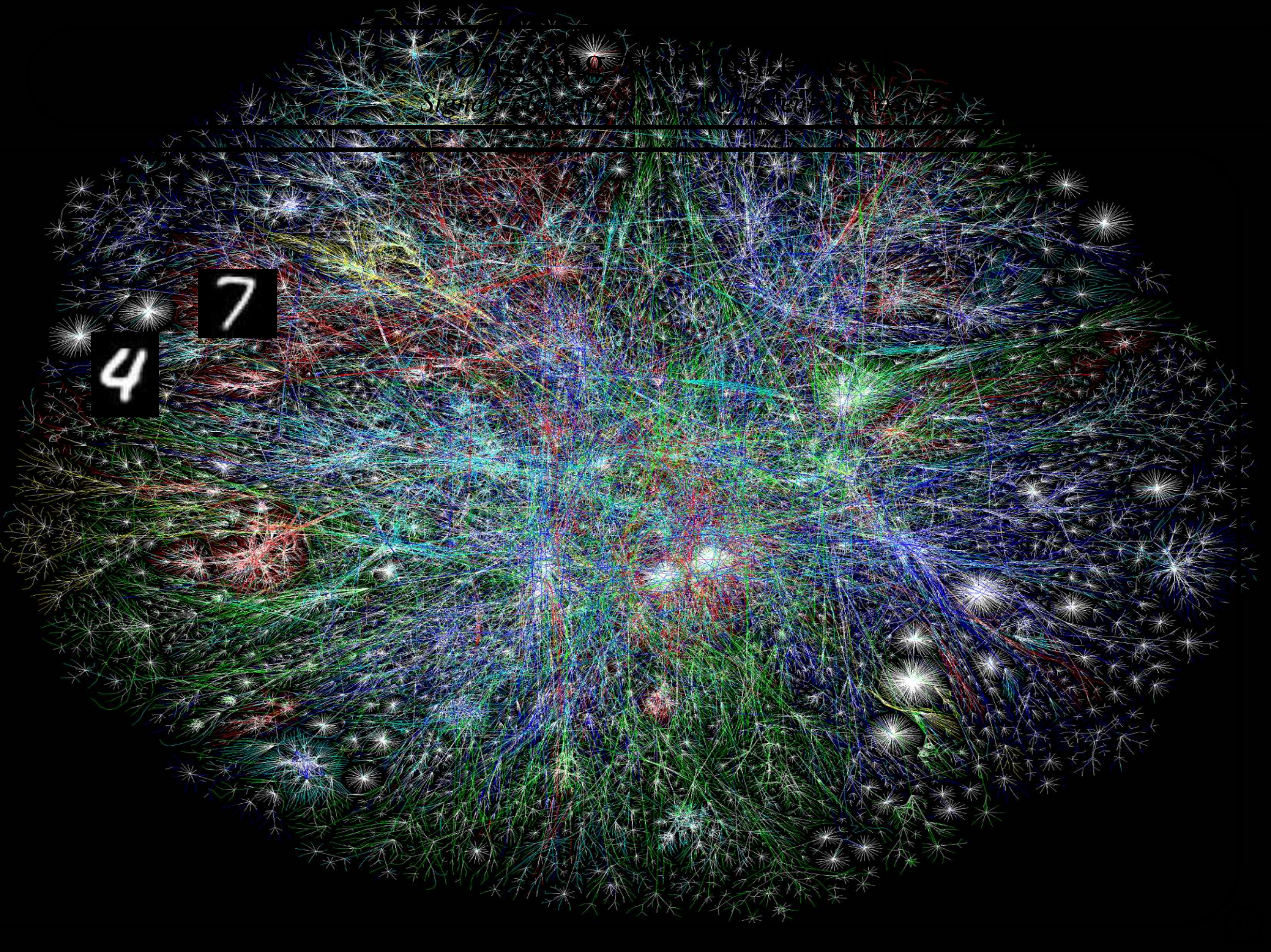
G_1

G_2

Signature

4

7

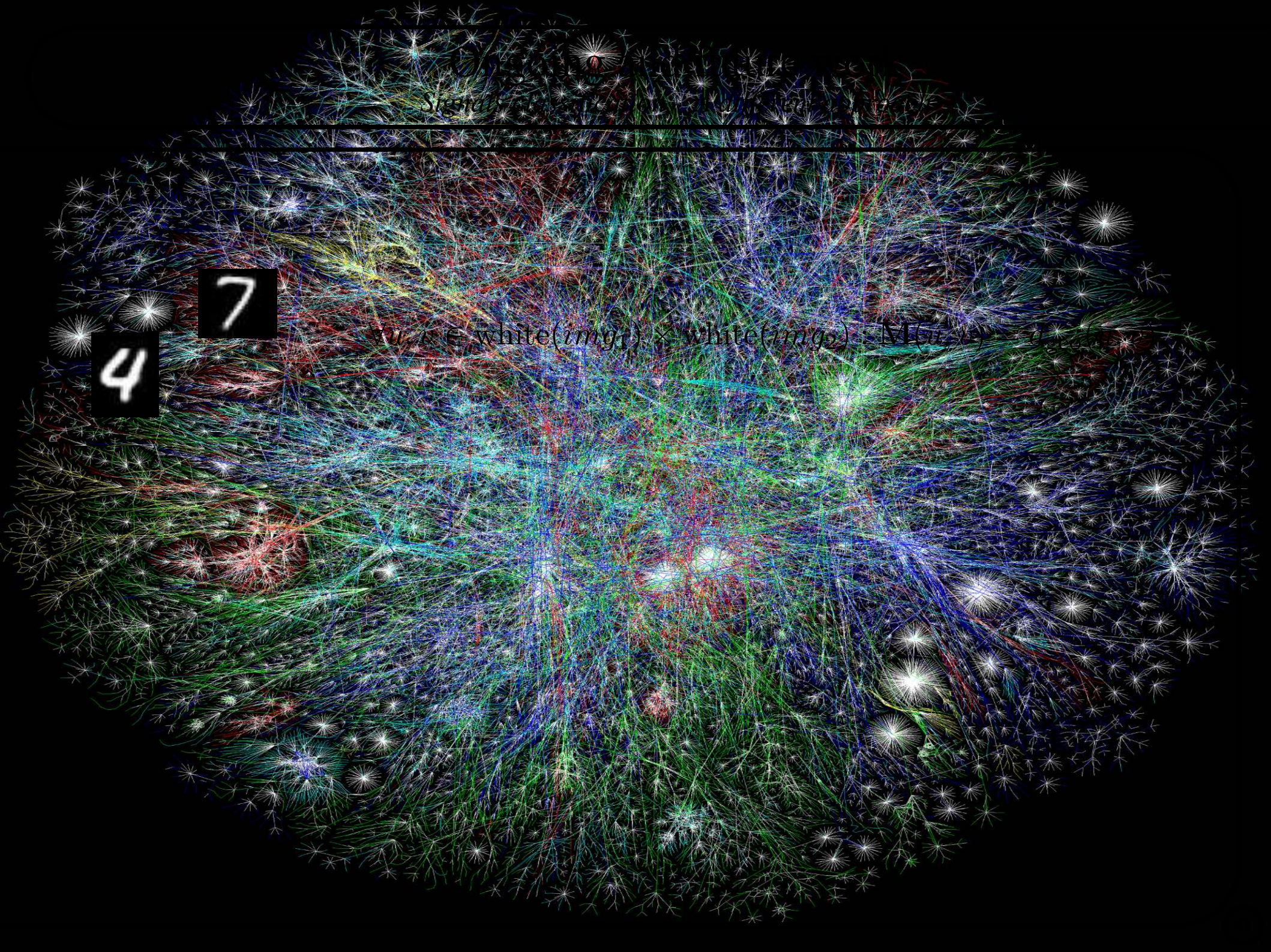


$\mathcal{O}(\text{white}(img_1) \times \text{white}(img_2))$
 $\text{Sigma}(\text{white}(img_1) \times \text{white}(img_2))$

4

7

$\forall u, v \in \text{white}(img_1) \times \text{white}(img_2) : M(u, v) = d_{\text{edge}}(u, v)$



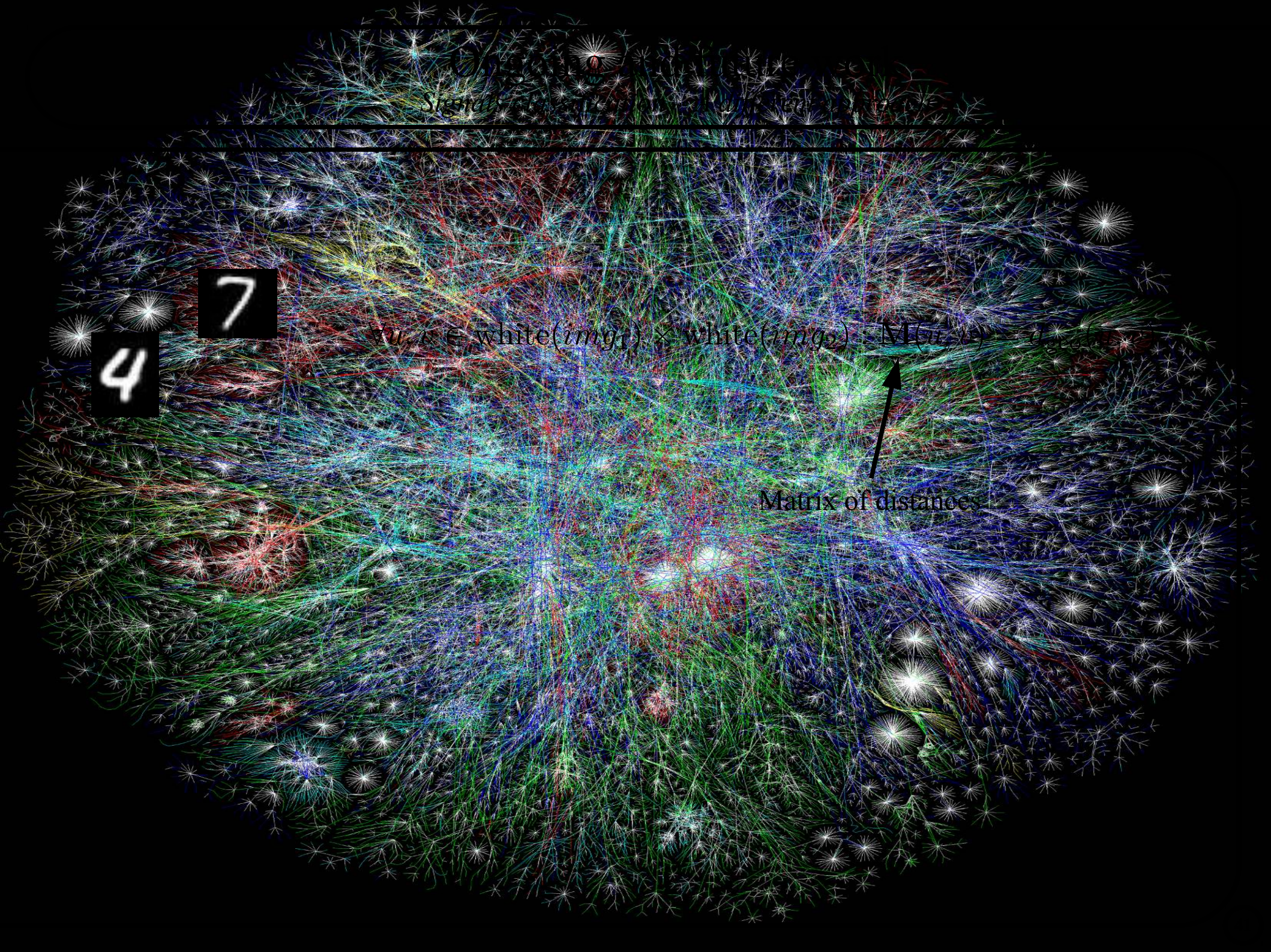
On the left
Signed distance from the left

4

7

$\forall u, v \in \text{white}(img_1) \times \text{white}(img_2) : M(u, v) = d(u, v)$

Matrix of distances



On the left
Signed distance to the nearest white pixel

4

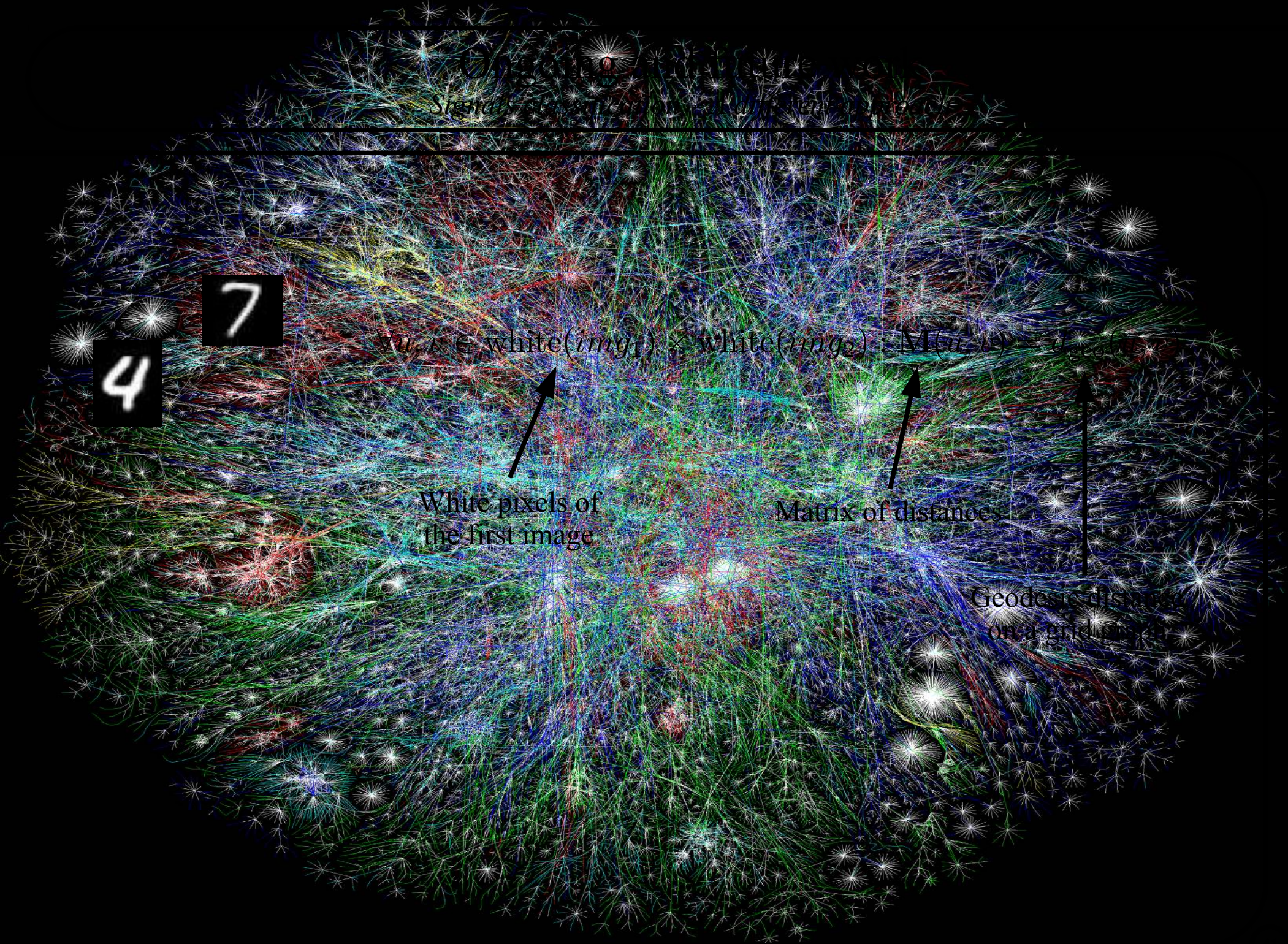
7

$\forall u, v \in \text{white}(img_1) \times \text{white}(img_2) : M(u, v) = d_{\text{grid}}(u, v)$

Matrix of distances

Geodesic distance
on a grid of pixels

Optimal Signature Extraction



Optimal
Signature

4

7

$$\forall u, v \in \text{white}(img_1) \times \text{white}(img_2) : M(u, v) = d_1 + d_2$$

$$d_1 = |\# \text{white}(img_1) - \# \text{white}(img_2)|$$

Hungarian
method

d_2 : Minimum cost to go
from img_1 to img_2



 Sigmond et al. (2012)

4

7

$$\forall u, v \in \text{white}(img_1) \times \text{white}(img_2) : M(u, v) = d_{\text{edge}}(u, v)$$

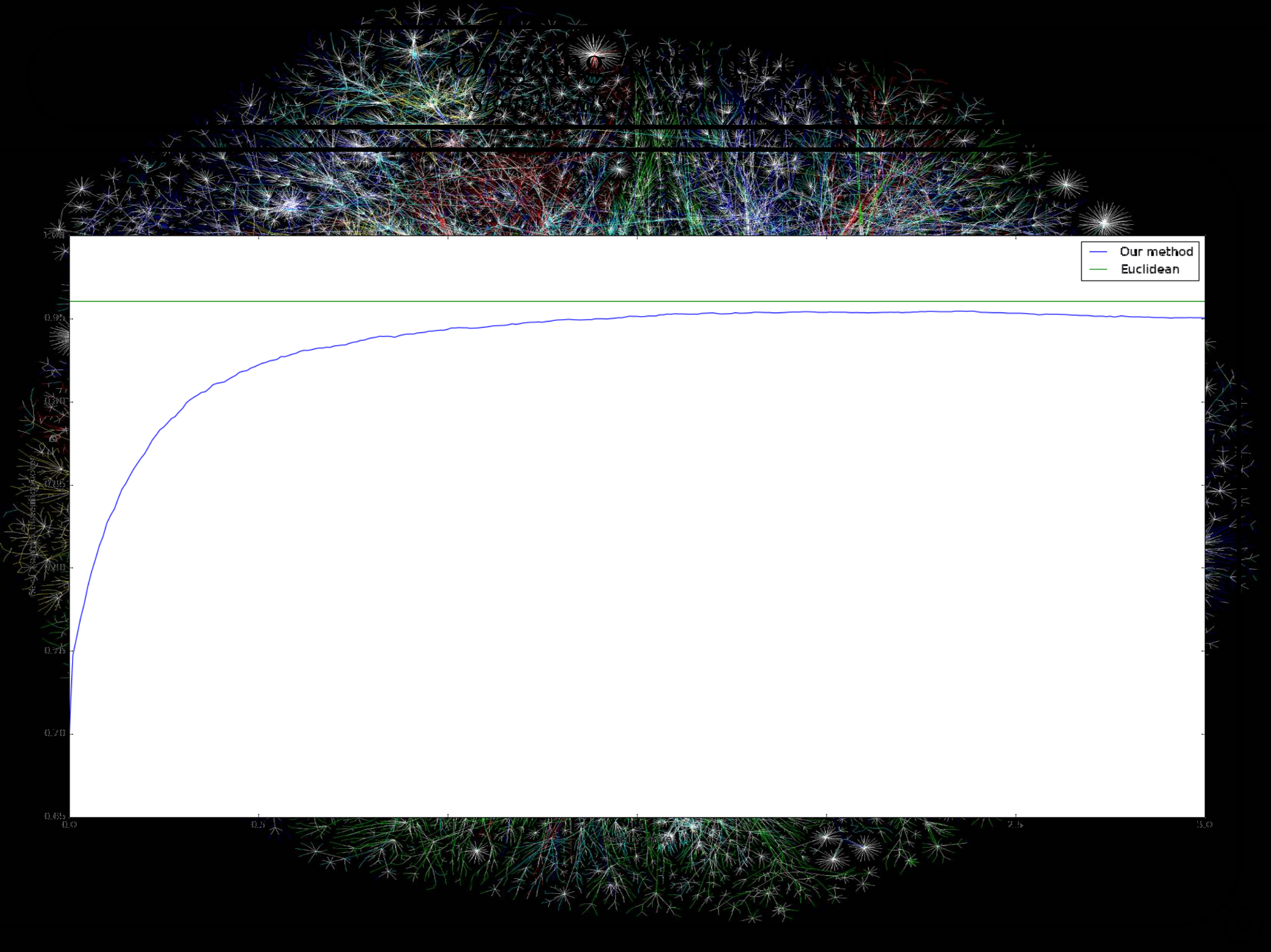
$$d_1 = |\# \text{white}(img_1) - \# \text{white}(img_2)|$$

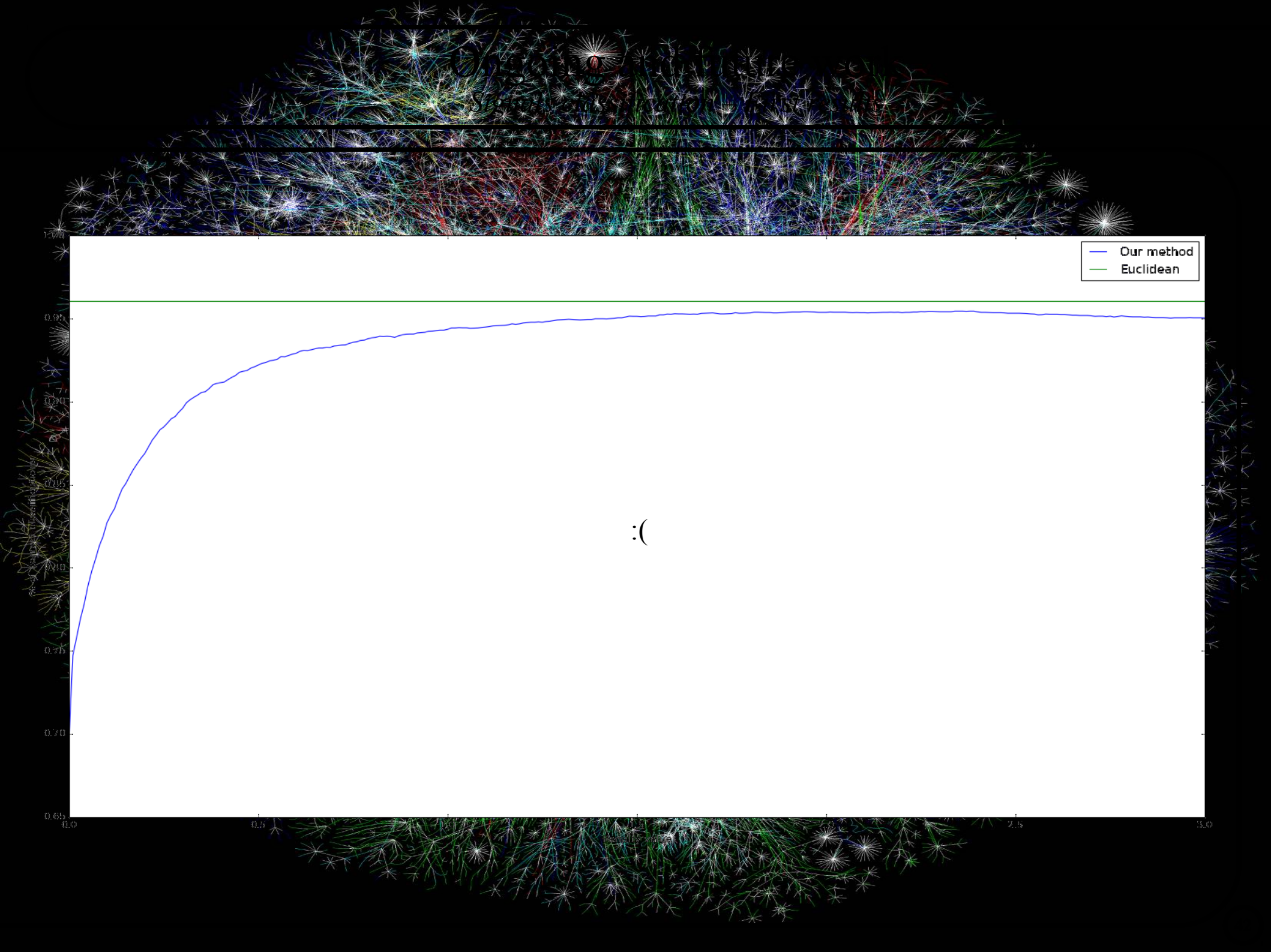
Hungarian method

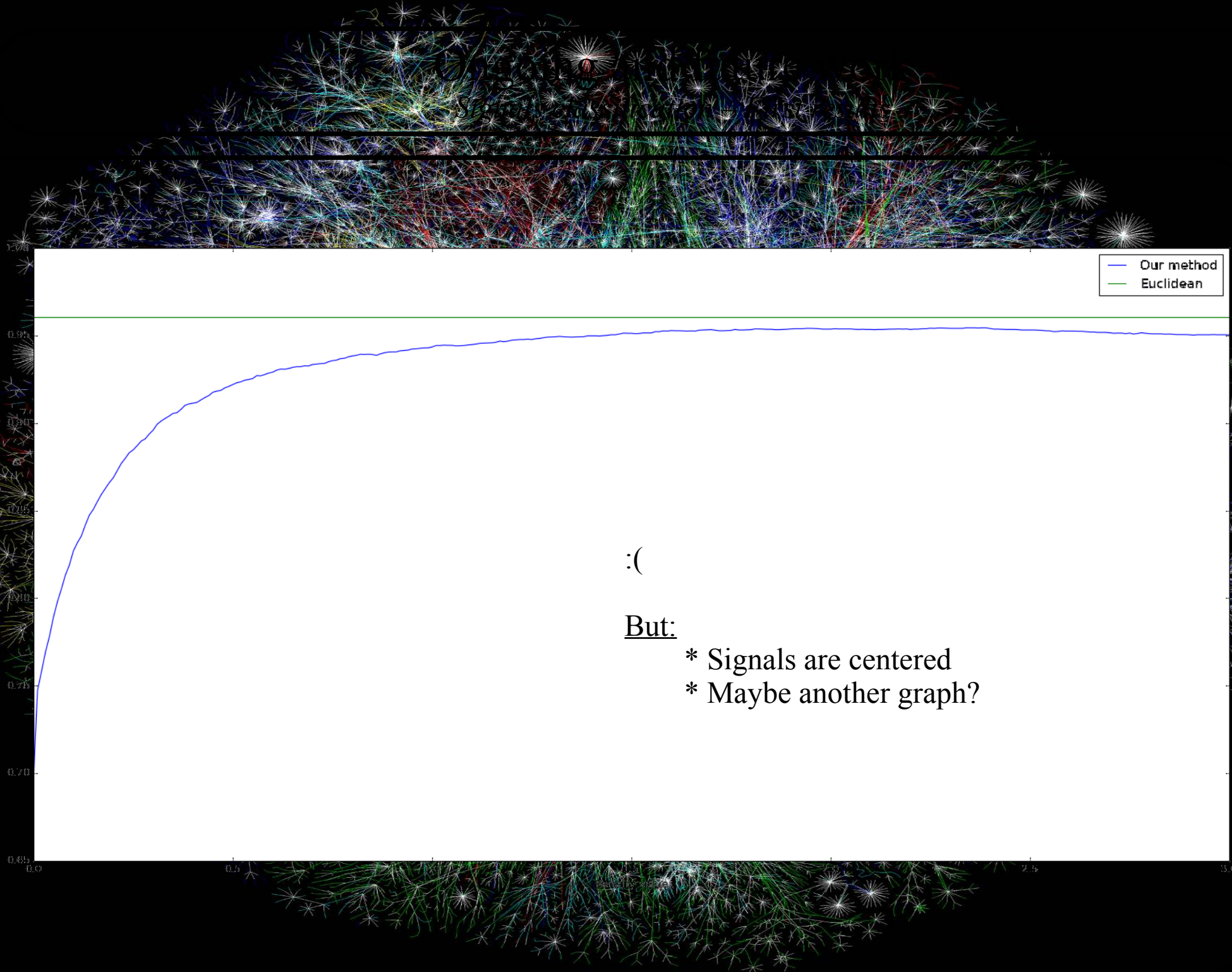
$$K\text{-NN}$$

$$d_1 + \alpha d_2$$

d_2 : Minimum cost to go from img_1 to img_2







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Characterization and Detection of Processes from Observations of Stationary

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Dominique Pastor*, Michael G. Rabbat*

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